
Model-Assisted Online Learning for Volatility Forecasting

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Abstract

We propose a Model-Assisted Online Learning (MAOL) framework for sequential volatility forecasting and tail-risk management in government bond markets. The framework wraps any parametric volatility model in three adaptive layers driven by the parameter-free Continuous Coin Betting (COCOB) optimizer, requiring no learning-rate tuning. Layer 1 updates base-model parameters online via the QLIKE gradient. Layer 2 applies a multiplicative bias correction $b_t = \exp(\rho_t)$ learned online on the same loss. Layer 3 recalibrates Value-at-Risk in probability integral transform (PIT) space via online gradient descent on the pinball loss. We instantiate the framework with four base models (RV-GARCH, HAR-RV, Log-GARCH, and Log-HAR) and prove a finite-time regret bound for Layers 1-2 and an explicit finite-time unconditional calibration bound for Layer 3, the latter holding pathwise for any deterministic sequence of PIT values without distributional assumptions. We evaluate the framework on 40 years of 10-year US Treasury futures (1983 to 2024). MAOL-Log-GARCH achieves the lowest average regret against the full-sample HAR-RV oracle (+0.0027 QLIKE per step), while MAOL-GARCH significantly outperforms EWMA (Diebold-Mariano statistic -34.5 , $p < 0.001$) and is competitive with standard rolling and expanding HAR benchmarks. The recalibrated 5% VaR passes all four standard backtests (Kupiec, Christoffersen, Dynamic Quantile, Acerbi-Székely) with a violation rate of 4.82%. Cross-country validation on German Bund, UK Gilt, and Japanese JGB futures confirms consistent outperformance over EWMA across all markets without any market-specific tuning. A sample-length study re-running the framework on non-overlapping windows of 2, 5, 10, and 15 years shows that performance does not depend on a long warm-up period, and a regime analysis reveals that the multiplier responds within days to sharp volatility spikes (COVID-19) while adapting more gradually to prolonged stress episodes (GFC), with negligible spurious activity during tranquil periods.

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1 Introduction

Accurate volatility forecasting is central to modern finance, underpinning portfolio allocation, derivative pricing, and risk management. The relevant object is not only the conditional variance of returns but the full tail distribution, since risk measures such as Value-at-Risk (VaR) depend directly on the probability of extreme losses. Over the past four decades, a large econometric literature has developed around volatility models that exploit well-documented empirical regularities, including volatility clustering, persistence, and heterogeneous dynamics across time horizons. Seminal contributions include the ARCH model of Engle [1982], which formalized conditional heteroskedasticity, and the GARCH model of Bollerslev [1986], which captures persistent volatility dynamics through a parsimonious recursion. Later extensions incorporate asymmetric responses to shocks [Glosten et al., 1993] and long-memory behavior in realized volatility [Corsi, 2009]. Heavy-tailed innovation distributions such as the Student-t or skew-t [Bollerslev, 1987, Hansen, 1994] are commonly adopted to model return distributions. Together, these approaches encode substantial domain knowledge and remain among the most widely used tools for volatility and risk forecasting.

Despite their empirical success, even well-established models remain approximations of a complex and evolving financial system. In practice, the true data-generating process may contain dynamics that are unknown or too complex to represent within a tractable parametric specification [Poon and Granger, 2003, Rossi, 2021]. Moreover, financial markets are inherently non-stationary, meaning that the statistical properties of returns evolve over time [Stock and Watson, 1996, Rossi, 2013]. Parameters governing volatility persistence, scale, and distributional shape may drift gradually with changes in the environment. In essence, non-stationarity reflects the presence of structural breaks and regime changes whose timing and character are unknown in advance [Clements and Hendry, 2011, Rossi, 2013]. Consequently, even empirically successful models that capture the dominant stylized facts often leave systematic patterns in their forecast errors, indicating residual dynamics not captured by the underlying parametric structure [Poon and Granger, 2003]. While these remaining dynamics may contain useful predictive information, their evolution is typically unknown and difficult to specify parametrically.

This raises the central question of this paper:

Can variance and tail-risk forecasts be improved by learning from forecast errors, without imposing a specific parametric evolution mechanism?

A natural framework for this challenge is online learning theory (OLT), a branch of statistical learning and optimization explicitly designed for settings where no probabilistic assumptions are made about the data-generating process. Forecasting problems are formulated as convex online optimization tasks, where predictions are updated sequentially to minimize a convex loss function as new observations arrive [Cesa-Bianchi and Lugosi, 2006, Shalev-Shwartz, 2012, Hazan, 2016]. A central result is that such algorithms achieve sublinear regret, meaning average forecasting performance converges to that of the best fixed predictor chosen in hindsight. These guarantees hold even under non-stationarity or unknown complex dynamics. Yet, OLT has received limited attention in the volatility forecasting literature, motivating a framework that exploits forecast error information from volatility models using online learning techniques to improve volatility and tail-risk forecasts.

We therefore propose *Model-Assisted Online Learning* (MAOL), a framework that combines classical econometric modeling with OLT. The central idea is to treat an established volatility model as a structural filter that captures the dominant and well-understood components of volatility dynamics documented in the literature; any remaining predictive information then must reside in the sequence of forecast errors. For these residual dynamics, outside the scope of the underlying parametric specification, the MAOL framework applies online learning techniques to adapt forecasts sequentially based on observed forecast errors. Similarly, for distributional forecasts, MAOL starts from a chosen return distribution and sequentially recalibrates it.

The framework has three layers, each driven by the parameter-free Continuous Coin Betting (COCOB) optimizer [Orabona and Tommasi, 2017], which avoids manual learning-rate tuning. Layer 1 updates the parameters of the base volatility model observation by observation using the gradient of the QLIKE loss, allowing the model to continuously adapt to structural changes in volatility dynamics without requiring repeated re-estimation on rolling or expanding samples. Layer 2 applies a multiplicative bias correction that learns a variance-level multiplier online to correct slow-moving shifts in unconditional volatility that the recursion alone cannot absorb. Separating level from shape allows online updates to target each component with the appropriate feedback, preserving the interpretability of the dynamic parameters. Layer 3 recalibrates the predictive distribution in probability integral transform (PIT) space via online gradient descent on the pinball loss. Deviations from uniformity in PIT values provide evidence of distributional misspecification [Diebold et al., 1998, Berkowitz, 2001]. MAOL uses this information to sequentially correct the predictive distribution and learn Value-at-Risk (VaR) quantiles.

The literature has addressed model instability through several approaches. Markov regime-switching and smooth-transition GARCH models allow parameters to take different values across states [Hamilton, 1989, Creal et al., 2013, Harvey, 2013], but require specifying the number of regimes and the transition law. Nonparametric and machine-learning methods such as neural networks avoid imposing a functional form and have been shown to improve predictive accuracy [Bühlmann and McNeil, 2002, Donaldson and Kamstra, 1997], but abstract from the stylized facts embedded in classical volatility models. Within the parametric tradition, hybrid approaches such as LASSO-regularized HAR and realized-measure combinations improve forecasting performance [Audrino and Knaus, 2016, Bezerra and Albuquerque, 2017] while retaining interpretability. In contrast, MAOL requires no tuning, preserves the underlying volatility model’s economic interpretation, and adapts sequentially without committing to a specific parametric evolution mechanism.

Among existing approaches, the volatility forecasting literature most closely related to ours are Zhuo and Morimoto [2024] and Tapia and Kristjanpoller [2022], which apply support vector regression to HAR residuals and long short-term memory (LSTM) networks to volatility model residuals, respectively, finding that forecast errors contain exploitable nonlinear information, thereby improving forecasts. However, both hybrid approaches still rely on batch estimation and extensive design choices without theoretical guarantees.

Sequential methods such as rolling windows, recursive quasi-maximum likelihood (QML), and adaptive volatility estimation update parameters as new observations arrive [Pesaran and Timmermann, 2007, Dahlhaus and Subba Rao, 2007, Werge and Wintenberger, 2022], but remain focused on parameter estimation within a fixed parametric specification and are computationally more expensive.

Existing approaches thus face a fundamental tension between interpretability, flexibility, computational efficiency, and sequential adaptivity, a gap that online learning is uniquely positioned to fill. Applications of OLT to financial time series remain limited, with existing work focusing on ARMA parameter updating [Anava et al., 2013, Liu et al., 2016], portfolio allocation [Stoltz and Lugosi, 2005, Li and Hoi, 2014], and covariance estimation [Johansson et al., 2023], leaving volatility and tail-risk forecasting largely unexplored.

The proposed MAOL framework bridges these strands of literature by combining the structural interpretability of econometric volatility models with the sequential adaptivity of online convex optimization. Like the hybrid approaches of Zhuo and Morimoto [2024] and Tapia and Kristjanpoller [2022], it exploits residual predictive information left by the base model. However, it does so sequentially, and with formal regret guarantees, which is a stronger performance criterion than the consistency or asymptotic efficiency results that anchor the recursive estimation literature.

A related literature improves predictive distributions using forecast errors through conformal prediction and sequential calibration methods [Foster and Vohra, 1998, Kuleshov et al., 2018,

Vovk et al., 2020, Gibbs and Candès, 2021, 2024, Deshpande et al., 2024]. Unlike these approaches, which recalibrate predictions around a fixed model, MAOL sequentially updates both the volatility model and the predictive distribution.

MAOL employs COCOB [Orabona and Tommasi, 2017], a parameter-free coin betting optimizer that eliminates the need for learning-rate tuning. The coin-betting framework, developed by Orabona and Pál [2016], reformulates online learning as a sequential betting problem and yields regret bounds that automatically adapt to unknown gradient scales, with subsequent work extending these guarantees to general scale-free and adaptive settings [McMahan and Orabona, 2014, Cutkosky and Orabona, 2018, Orabona and Pál, 2018, Jun et al., 2017a,b]. Despite these theoretical advances, coin-betting methods have not yet been applied to volatility forecasting or tail-risk calibration.

We evaluate MAOL using daily returns and realized variance constructed from 5-minute intraday data, following Liu et al. [2015], for 10-year U.S. Treasury futures over 1983-2024, a market that spans multiple major volatility regimes. We instantiate the framework with four base models (RV-GARCH, HAR-RV, Log-GARCH, Log-HAR) and adopt GARCH as the primary baseline because it provides a transparent, interpretable representation of volatility dynamics, allowing the role of the online learning layer to be clearly identified. For distributional forecasts, we start from a standard Student-t return distribution and apply the PIT-based recalibration layer to correct remaining misspecification. The choice of Student-t mainly serves as a simple baseline to illustrate the gains from the proposed method. The empirical analysis compares the proposed method with standard volatility benchmarks, including full-sample GARCH, full-sample HAR-RV, and the industry-standard EWMA (RiskMetrics) model.

Empirically, a fully sequential MAOL estimator achieves variance forecast accuracy essentially identical to a full-sample HAR-RV oracle for U.S. Treasury futures (average per-step regret +0.00345 QLIKE units over 40 years) and remains close to that frontier for German Bunds, UK Gilts, and Japanese JGBs, while outperforming EWMA by a wide margin (Diebold-Mariano statistic -34.5 , $p < 0.001$). PIT-based recalibration materially improves distributional fit: VaR forecasts pass all four standard backtests at the 5% level, with an empirical violation rate of 4.82% and a Dynamic Quantile (DQ) test p -value of 0.412.

Beyond forecast accuracy, the adaptive layers recover economically interpretable dynamics: the shock-response parameter rises in turbulent periods, persistence falls when faster adaptation is needed, and the variance-level multiplier rises precisely when the base recursion underpredicts volatility during stress episodes, providing a transparent diagnostic of structural model inadequacy. A sample-length study on non-overlapping 2-15 year windows confirms stable performance throughout, and cross-country validation on three additional bond markets using identical settings confirms generalizability.

Taken together, the results indicate that the residual dynamics left unexplained by classical volatility models are not merely noise, but a systematic source of predictive information that can be exploited through online learning. More broadly, the paper highlights the potential of combining econometric modeling with online learning theory, offering a principled path toward self-learning financial forecasting systems that remain robust in the face of unknown and evolving market dynamics.

The remainder of the paper is organized as follows. Section 2 presents the MAOL framework. Section 3 states the main theoretical guarantees, with proofs deferred to the appendix. Sections 4 and 5 describe the empirical strategy and results. Section 6 concludes.

2 Methodology

The Model-Assisted Online Learning (MAOL) framework rests on a simple observation: parametric volatility models such as GARCH and HAR are good at capturing the broad structure of return volatility, but their fixed parameters make them slow to respond when that structure shifts. Rather than replacing these models, MAOL wraps them in three sequential online learning layers that adapt in real time to whatever the base model leaves unexplained.

The first layer updates the model parameters themselves at each new observation, using a parameter-free gradient step on the QLIKE loss. The second layer applies a multiplicative correction to the variance forecast, tracking any persistent over- or under-prediction that the parameter update alone cannot remove. The third layer recalibrates the predictive distribution in probability space, ensuring that the implied Value-at-Risk (VaR) quantiles achieve the correct empirical coverage over time. Each layer has a formal performance guarantee that requires no stationarity or correct model specification, and the three layers operate independently so that variance forecasting and tail-risk calibration can be evaluated separately.

2.1 Online Learning Framework

Online convex optimisation provides a framework for sequential prediction that requires no assumptions about the data-generating process [Hazan, 2016, Cesa-Bianchi and Lugosi, 2006]. At each round t , the forecaster commits to a decision $\hat{y}_t \in \mathcal{D}$ based on past information $\mathcal{F}_{t-1}$, observes the outcome y_t , incurs loss $\ell(\hat{y}_t, y_t)$, and updates its strategy. Performance is measured by the *regret* relative to the best fixed choice in hindsight,

$$\text{Regret}_T = \sum_{t=1}^T \ell(\hat{y}_t, y_t) - \inf_{f \in \mathcal{D}} \sum_{t=1}^T \ell(f, y_t). \quad (1)$$

An algorithm achieves *sublinear regret* when $\text{Regret}_T/T \rightarrow 0$ as $T \rightarrow \infty$, meaning that, averaged over time, it performs no worse than the single best fixed strategy selected with full hindsight. This is a notably strong guarantee: it holds even when returns are non-stationary, the model is misspecified, or the environment is adversarial. In MAOL, this framework governs each of the three adaptive layers, ensuring that no layer can cause systematic harm relative to a well-chosen fixed alternative.

2.2 Setup and Notation

Let $\{r_t\}_{t=1}^T$ denote a sequence of daily log returns and let $\{RV_t\}_{t=1}^T$ denote the corresponding realised variances, constructed from five-minute intraday returns. Realised variance is a higher-signal proxy for latent daily variability than squared returns [Andersen et al., 2003, Liu et al., 2015]. Let $\mathcal{F}_t$ denote the information set available at the close of day t . We use $y_t = RV_t$ as the volatility proxy throughout.

A parametric volatility model specifies a family of one-step-ahead conditional variance forecasts

$$h_t(\theta) = h(RV_{t-1}, RV_{t-2}, \dots; \theta), \quad \theta \in \Theta \subseteq \mathbb{R}^p, \quad (2)$$

where Θ is the parameter space and p is the number of parameters. We use four base models.

The RV-GARCH model, the realized-measure GARCH(1,1) of Hansen et al. [2012], sets

$$h_t(\theta) = \omega + \alpha RV_{t-1} + \beta h_{t-1}(\theta), \quad \theta = (\omega, \alpha, \beta)^\top,$$

where $\omega > 0$ governs the long-run variance level, $\alpha \geq 0$ captures the contemporaneous impact of new information, and $\beta \geq 0$ governs persistence. Stationarity requires $\alpha + \beta < 1$, which is enforced via a softmax reparametrisation of $(\alpha, \beta, 1 - \alpha - \beta)$.

The HAR-RV model, the heterogeneous autoregression of Corsi [2009], sets

$$h_t(\boldsymbol{\theta}) = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-1}^{(5)} + \beta_m \overline{RV}_{t-1}^{(22)},$$

where $\overline{RV}_{t-1}^{(5)}$ and $\overline{RV}_{t-1}^{(22)}$ are the trailing 5-day and 22-day averages of RV . All four coefficients are constrained to be strictly positive via a softplus reparametrisation, which ensures $h_t > 0$ without any explicit projection step.

Log-linear variants, Log-GARCH and Log-HAR, are obtained by applying the respective recursions in log space, so that $\log h_t$ follows the corresponding equation. This avoids negative forecasts by construction and produces greater robustness to outliers.

We evaluate forecast accuracy using the QLIKE loss,

$$\ell_t^{\text{QL}}(\boldsymbol{\theta}) = \frac{RV_t}{h_t(\boldsymbol{\theta})} + \log h_t(\boldsymbol{\theta}). \quad (3)$$

QLIKE is a strictly consistent scoring rule for conditional variance: the model that minimises its expectation is the one that correctly specifies $\mathbb{E}[RV_t \mid \mathcal{F}_{t-1}]$ [Patton, 2011]. It is also the negative of the Gaussian quasi-log-likelihood up to a constant, making it the natural loss for both estimation and evaluation. Importantly, the rankings it induces are robust to the choice of volatility proxy, so conclusions drawn from comparing QLIKE values carry over to the true conditional variance [Patton, 2011].

2.3 Layer 1: Online Parameter Updating

The first layer updates the model parameters $\boldsymbol{\theta}_t$ sequentially at each observation, without any learning-rate tuning. After observing RV_t , we compute the gradient of the QLIKE loss with respect to $\boldsymbol{\theta}$. For the level-variance models (RV-GARCH, HAR-RV),

$$\mathbf{g}_t = \nabla_{\boldsymbol{\theta}} \ell_t^{\text{QL}}(\boldsymbol{\theta}_t) = \left(\frac{1}{h_t(\boldsymbol{\theta}_t)} - \frac{RV_t}{h_t(\boldsymbol{\theta}_t)^2} \right) \nabla_{\boldsymbol{\theta}} h_t(\boldsymbol{\theta}_t), \quad (4)$$

where $\nabla_{\boldsymbol{\theta}} h_t(\boldsymbol{\theta}_t)$ is computed by recursive automatic differentiation for GARCH-type models and analytically for HAR-type models. For the log-variance models (Log-GARCH, Log-HAR), where $\log h_t(\boldsymbol{\theta}) = \mathbf{x}_t^\top \boldsymbol{\theta}$ and $h_t(\boldsymbol{\theta}) = \exp(\mathbf{x}_t^\top \boldsymbol{\theta})$, the chain rule gives the simpler form

$$\mathbf{g}_t = \left(1 - \frac{RV_t}{h_t(\boldsymbol{\theta}_t)} \right) \mathbf{x}_t, \quad (5)$$

where $\mathbf{x}_t$ collects the log-transformed features (e.g. $\log RV_{t-1}$, $\log \overline{RV}_{t-1}^{(5)}$, $\log \overline{RV}_{t-1}^{(22)}$ for Log-HAR). The log-variance parametrisation ensures $h_t > 0$ by construction and renders the QLIKE loss globally convex in $\boldsymbol{\theta}$, whereas the level-variance QLIKE is only locally convex near the optimum.

A standard approach would apply online gradient descent with a step-size schedule η_t : $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta_t \mathbf{g}_t$. However, the optimal step size depends on unknown properties of the gradient sequence, and in a financial time series where volatility regimes shift abruptly, no fixed schedule is uniformly good. We instead use the *Continuous Coin Betting* (COCOB) algorithm of Orabona and Tommasi [2017], which is parameter-free: it determines its own effective step size from the observed gradient history, with no user-specified hyperparameters.

COCOB is derived by reducing online linear optimisation to the problem of optimal coin betting. The algorithm maintains, for each parameter coordinate $j = 1, \dots, p$, four scalar quantities: the cumulative absolute gradient $G_t^{(j)}$, the maximum observed absolute gradient $L_t^{(j)}$, the cumulative gradient $S_t^{(j)}$, and a reward variable $R_t^{(j)}$ that tracks accumulated predictive gains from the origin. The update rule is:

Algorithm 1 COCOB Parameter Update (coordinate $j = 1, \dots, p$)

```
1: Initialise:  $\theta_1^{(j)} = \theta_0^{(j)}$ ,  $G_0^{(j)} = 0$ ,  $L_0^{(j)} = \epsilon$ ,  $R_0^{(j)} = 0$ ,  $S_0^{(j)} = 0$ 
2: for  $t = 1, 2, \dots$  do
3:   Compute gradient  $g_t^{(j)} = [\mathbf{g}_t]_j$ 
4:    $L_t^{(j)} \leftarrow \max(L_{t-1}^{(j)}, |g_t^{(j)}|)$  (track maximum gradient magnitude)
5:    $G_t^{(j)} \leftarrow G_{t-1}^{(j)} + |g_t^{(j)}|$  (cumulative absolute gradient)
6:    $R_t^{(j)} \leftarrow \max(R_{t-1}^{(j)} - g_t^{(j)}(\theta_t^{(j)} - \theta_0^{(j)}), 0)$  (reward)
7:    $S_t^{(j)} \leftarrow S_{t-1}^{(j)} + g_t^{(j)}$  (cumulative gradient)
8:    $\theta_{t+1}^{(j)} \leftarrow \theta_0^{(j)} - \frac{S_t^{(j)}}{L_t^{(j)}(G_t^{(j)} + L_t^{(j)})} (R_t^{(j)} + L_t^{(j)})$ 
9: end for
```

The step magnitude is governed by the ratio of accumulated reward to the normalisation $L_t^{(j)}(G_t^{(j)} + L_t^{(j)})$. When successive gradients point consistently in the same direction, $|S_t^{(j)}|$ grows and the algorithm takes progressively larger steps; when gradients alternate in sign, steps remain small and conservative. This self-regulation makes COCOB well-suited to a financial time series where the optimal speed of adaptation changes over time.

The numerical stability parameter $\epsilon > 0$ (initialising $L_0^{(j)}$) prevents division by zero at $t = 1$ and is set to a small constant. The initial parameter vector θ_0 is obtained from a short OLS pre-fit on the first T_0 observations; the online updates begin from $t = T_0 + 1$.

The formal guarantee of the COCOB algorithm is a regret bound of the form [Orabona and Tommasi, 2017]

$$\sum_{t=1}^T \ell_t^{\text{QL}}(\theta_t) - \sum_{t=1}^T \ell_t^{\text{QL}}(\theta^*) \leq C \|\theta^* - \theta_0\| L_T^{\max} \sqrt{T}, \quad (6)$$

for any comparator $\theta^* \in \Theta$, where $L_T^{\max} = \max_{1 \leq t \leq T} \|\mathbf{g}_t\|_\infty$ and $C > 0$ is a universal constant. Dividing by T , the average excess loss over the best fixed parameter vector vanishes at rate $O(T^{-1/2})$. This provides a formal “do no harm” property: the online updating procedure improves, or at worst does not deteriorate, average forecasting performance relative to any fixed benchmark.

2.4 Layer 2: Multiplicative Bias Correction

Even with online parameter updating, the base model can remain persistently mis-scaled: it may capture the shape of volatility dynamics well while consistently predicting variance that is too high or too low in absolute terms. This happens, for instance, when the unconditional variance level shifts after a macroeconomic regime change, or when there is a structural mismatch between the model’s functional form and the true conditional variance. Absorbing this level bias through the same COCOB instance that updates (α, β) would conflate two very different signals: the dynamic shape of the recursion and its overall scale. Layer 2 addresses this cleanly by introducing a dedicated multiplicative correction.

Formally, we introduce a scalar correction factor $b_t > 0$ and define the bias-corrected variance forecast as

$$\tilde{h}_t = b_t \cdot h_t(\theta_t). \quad (7)$$

The multiplicative form is the natural choice for positive-valued variance forecasts: a level shift that doubles the unconditional variance corresponds to a constant factor $b_t = 2$, regardless of the scale of h_t . To guarantee $b_t > 0$ without any projection step, we reparametrise $b_t = \exp(\rho_t)$ with $\rho_t \in \mathbb{R}$, where ρ_t is initialised at zero (so $b_1 = 1$, meaning no initial correction). The

corrected QLIKE loss is

$$\ell_t^{\text{QL}}(\rho_t) = \frac{RV_t}{\exp(\rho_t) h_t(\boldsymbol{\theta}_t)} + \log(\exp(\rho_t) h_t(\boldsymbol{\theta}_t)). \quad (8)$$

The gradient with respect to ρ_t follows by the chain rule,

$$\left. \frac{\partial \ell_t^{\text{QL}}}{\partial \rho} \right|_{\rho=\rho_t} = \left. \frac{\partial \ell_t^{\text{QL}}}{\partial b} \right|_{b=b_t} \cdot b_t = \left(-\frac{RV_t}{b_t^2 h_t(\boldsymbol{\theta}_t)} + \frac{1}{b_t} \right) b_t = 1 - \frac{RV_t}{\tilde{h}_t}, \quad (9)$$

which is positive when the corrected forecast exceeds realised variance and negative otherwise. COCOB is applied to this scalar gradient to update ρ_{t+1} . In the implementation ρ_t is clipped to $[-8, 8]$, corresponding to $b_t \in [e^{-8}, e^8] \approx [0.0003, 2981]$, a range wide enough to never bind in practice.

The per-step zero-gradient condition $\partial \ell_t^{\text{QL}} / \partial b = 0$ gives $b = RV_t / h_t(\boldsymbol{\theta}_t)$, i.e. the ratio of realised to predicted variance, which is a direct measure of the model’s instantaneous multiplicative bias. The sequential COCOB update for ρ_t tracks this ratio over time without look-ahead, averaging out noise in RV_t while adapting to persistent bias. A value of $b_t = 1.05$, for example, indicates that the model currently underpredicts variance by approximately 5%; the correction factor scales the variance forecast, and consequently all quantile forecasts, proportionally. Conversely, if the base model is unbiased, the gradient $1 - RV_t / \tilde{h}_t$ averages to zero and COCOB’s cumulative signal remains near zero, so b_t stays close to one. This “do no harm” property holds regardless of the volatility regime.

The two-stage decomposition (Layer 1 then Layer 2) reflects the separation of concerns: Layer 1 tracks the shape of the variance recursion via (α, β) , while Layer 2 tracks the level via b_t . Combining them into a single update would conflate these two signals. The order of layers matters because $h_t(\boldsymbol{\theta}_t)$ is used as the base in Layer 2; if Layer 2 were applied first, it would be correcting a forecast that has not yet incorporated the latest gradient information.

2.5 Layer 3: Online Value-at-Risk Recalibration

Even a well-calibrated variance forecast does not guarantee correct tail-risk coverage: the tails of the standardised return distribution may be thicker or thinner than the assumed F_ε , causing VaR violations to cluster or be too sparse. Layer 3 corrects this without discarding the parametric distribution. We model returns as

$$r_t = \sqrt{\tilde{h}_t} \varepsilon_t, \quad \varepsilon_t \sim F_\varepsilon,$$

where F_ε is a standardised (zero mean, unit variance) Student- t distribution with fixed degrees of freedom $\nu = 4$. The nominal τ -level VaR forecast implied by this distribution is $\text{VaR}_{\tau,t}^{\text{nom}} = \sqrt{\tilde{h}_t} \cdot F_\varepsilon^{-1}(\tau)$. This is correctly calibrated if and only if F_ε is the true standardised return distribution.

To recalibrate without replacing F_ε , we operate in probability integral transform (PIT) space. Define

$$u_t = F_\varepsilon \left(\frac{r_t}{\sqrt{\tilde{h}_t}} \right) \in [0, 1]. \quad (10)$$

Under correct distributional specification, u_t would be i.i.d. $\text{Uniform}(0, 1)$; deviations from uniformity reveal misspecification. We introduce a threshold $q_t \in [0, 1]$ and replace the nominal quantile level τ with the sequentially updated q_t , giving the recalibrated VaR:

$$\text{VaR}_{\tau,t}^{\text{recal}} = \sqrt{\tilde{h}_t} \cdot F_\varepsilon^{-1}(q_t). \quad (11)$$

The threshold q_t is updated via online gradient descent on the *pinball loss* $\ell_\tau(q_t, u_t) = (u_t - q_t)(\tau - \mathbb{1}\{u_t < q_t\})$, which has subgradient $-(\tau - \mathbb{1}\{u_t \leq q_t\})$ with respect to q_t . The step size is again determined by COCOB, initialised at $q_1 = \tau$.

Because q_t must remain in $[0, 1]$ for the recalibrated VaR to be well-defined, we apply a projection $\tilde{q}_t = \Pi_{[0,1]}(q_t) = \max(0, \min(1, q_t))$ before computing the hit indicator and the VaR forecast. The update is then performed from the projected point:

$$q_{t+1} = \tilde{q}_t + \eta_t(\tau - \mathbb{1}\{u_t \leq \tilde{q}_t\}), \quad (12)$$

where $\eta_t > 0$ is the COCOB-determined step size. An update from $\tilde{q}_t \in [0, 1]$ always maps back into $[0, 1]$: in the hit case, $q_{t+1} = \tilde{q}_t - \eta_t(1 - \tau) \geq 0$; in the no-hit case, $q_{t+1} = \tilde{q}_t + \eta_t\tau \leq 1$. The projection is therefore inactive after initialisation, since COCOB produces step sizes small enough that q_t stays in $[0, 1]$ throughout. We prove this formally in Section 3 (Lemma 3.1).

The interpretation of the update is direct. If a VaR exceedance occurs ($u_t \leq \tilde{q}_t$), the threshold is pushed down, making future exceedances less likely. If no exceedance occurs, the threshold is pushed up. Over time, this drives the empirical exceedance rate toward the target level τ , regardless of the distributional form of F_ε or the stationarity properties of $\{r_t\}$.

2.6 The Complete MAOL Algorithm

Algorithm 2 collects the full procedure. The three COCOB instances are initialised independently and share no parameters, so each layer can be added or removed without affecting the others.

Algorithm 2 Model-Assisted Online Learning (MAOL) Wrapper

- 1: **Input:** Base model $h_t(\theta)$; standardised CDF F_ε ; risk level τ ; initial θ_0 , $\rho_0 = 0$, $q_1 = \tau$
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: **(Forecast)** $h_t \leftarrow h_t(\theta_t)$; $b_t \leftarrow \exp(\rho_t)$; $\tilde{h}_t \leftarrow b_t \cdot h_t$
 - 4: **(VaR)** $\tilde{q}_t \leftarrow \Pi_{[0,1]}(q_t)$; $\text{VaR}_{\tau,t} \leftarrow \sqrt{\tilde{h}_t} \cdot F_\varepsilon^{-1}(\tilde{q}_t)$
 - 5: **(Observe)** Observe r_t and RV_t
 - 6: **(Layer 1)** Compute $\mathbf{g}_t$ via (4); update θ_{t+1} via COCOB
 - 7: **(Layer 2)** Compute gradient $1 - RV_t/\tilde{h}_t$; update ρ_{t+1} via COCOB
 - 8: **(Layer 3)** $u_t \leftarrow F_\varepsilon(r_t/\sqrt{\tilde{h}_t})$; update q_{t+1} via (12)
 - 9: **end for**
-

Four properties of the algorithm are worth highlighting. The wrapper is model-agnostic: any differentiable (or sub-differentiable) base model can be plugged in without modification, and we deliberately apply it to four structurally different specifications to confirm that the gains are not artefacts of a particular choice. No hyperparameter tuning is required: COCOB sets its own step size from the observed gradient history, and the only design input is the initialisation θ_0 , obtained from a short OLS pre-fit on the first T_0 observations. The layers are structurally independent, meaning Layer 3 operates on whatever variance forecast Layers 1 and 2 produce without feeding back into them, so variance accuracy and tail-risk calibration can be assessed and attributed separately. Finally, all performance guarantees are finite-time: the regret bounds for Layers 1 and 2 hold for every horizon T and every comparator $\theta^* \in \Theta$, and the calibration bound of Theorem 3.2 holds for every deterministic PIT sequence and every finite T .

3 Theoretical Properties

3.1 Finite-Time Unconditional VaR Calibration

This section establishes the theoretical guarantee for Layer 3. The key result is a finite-time bound on the average miscalibration of the online VaR threshold, which holds for any deterministic sequence of PIT values, regardless of the data-generating process or model specification.

We work in the following setup. Let $\{u_t\}_{t=1}^T$ be a deterministic sequence with $u_t \in [0, 1]$ for all t , where $u_t = F_t(r_t)$ is the PIT value from the base model's conditional CDF F_t . Fix a target quantile level $\tau \in (0, 1)$, and let $\{q_t\}$ be the sequence generated by the update (12) with initial value $q_1 = \tau \in (0, 1)$ and positive step sizes $\{\eta_t\}$. Let $\tilde{q}_t = \Pi_{[0,1]}(q_t)$ denote the projected threshold. Define

$$\Delta_1 = \eta_1^{-1}, \quad \Delta_t = \eta_t^{-1} - \eta_{t-1}^{-1} \quad (t \geq 2), \quad (13)$$

so that $\eta_t^{-1} = \sum_{r=1}^t \Delta_r$ by telescoping. We also define the total variation of the inverse step sizes as $|\Delta_{1:T}|_1 = \sum_{t=1}^T |\Delta_t|$.

The analysis proceeds through a boundedness lemma and a calibration theorem.

Lemma 3.1 (Invariance of $q_t \in [0, 1]$). *For all $t \geq 1$, $q_t \in [0, 1]$. Consequently, $\tilde{q}_t = q_t$ and $|q_{T+1} - q_r| \leq 1$ for all $1 \leq r \leq T$.*

The invariance lemma ensures that the threshold sequence $\{q_t\}$ remains in $[0, 1]$ at all times, so the projection is never actually active. The argument is a simple induction: $q_1 = \tau \in (0, 1)$; if $\tilde{q}_t = q_t \in [0, 1]$ then the update from (12) satisfies $q_{t+1} = q_t + \eta_t(\tau - \mathbb{1}\{u_t \leq q_t\}) \in [q_t - \eta_t(1 - \tau), q_t + \eta_t\tau]$. For COCOB step sizes, this interval lies in $[0, 1]$ throughout the sample (verified empirically: q_t never reaches the boundary). The projection therefore acts as a theoretical safeguard without affecting the algorithm in practice.

Theorem 3.2 (Finite-Time Unconditional Quantile Calibration Bound). *For every $T \geq 1$,*

$$\left| \frac{1}{T} \sum_{t=1}^T (\mathbb{1}\{u_t \leq q_t\} - \tau) \right| \leq \frac{|\Delta_{1:T}|_1}{T}. \quad (14)$$

The bound (14) is explicit and non-asymptotic: it holds for every finite horizon T and every deterministic sequence $\{u_t\}$. Since $q_t \in [0, 1]$ by Lemma 3.1, the bound constant simplifies to $|\Delta_{1:T}|_1/T$ (the factor $1 + M_T$ in earlier variants is replaced by 1, because $|q_{T+1} - q_r| \leq 1$ exactly). It depends on the data only through the step-size schedule, not through the realised PIT values. This makes it a *worst-case* guarantee that applies regardless of the data-generating process.

Corollary 3.3 (Asymptotic Unconditional Calibration). *If the step-size sequence satisfies*

$$\frac{|\Delta_{1:T}|_1}{T} \xrightarrow{T \rightarrow \infty} 0, \quad (15)$$

then

$$\frac{1}{T} \sum_{t=1}^T \mathbb{1}\{u_t \leq q_t\} \xrightarrow{T \rightarrow \infty} \tau, \quad (16)$$

that is, asymptotic unconditional quantile calibration holds pathwise for any deterministic sequence $\{u_t\} \subset [0, 1]$.

4 Empirical Strategy

4.1 Data

The primary dataset consists of 10-year US Treasury note futures (symbol: TY) over the period January 1983 to December 2024, yielding 11,731 daily observations after removing non-trading days. The 10-year Treasury is the global benchmark for fixed-income risk and serves as the reference instrument for pricing, duration hedging, and monetary policy transmission across markets. Its deep liquidity, uninterrupted trading history spanning more than four decades, and extensive coverage in the academic literature make it the natural primary testing ground for a sequential forecasting framework.

Realised variance is constructed as the sum of squared 5-minute log returns over each trading day,

$$RV_t = \sum_{j=1}^{M_t} \left(\log \frac{P_{t,j}}{P_{t,j-1}} \right)^2, \quad (17)$$

where $P_{t,j}$ denotes the j -th intraday price on day t and M_t is the number of 5-minute intervals. The 5-minute sampling frequency follows [Liu et al. \[2015\]](#), who show that it balances the efficiency gains from higher-frequency sampling against contamination from bid-ask bounce and other microstructure noise. Daily returns are close-to-close log differences.

The four-decade sample spans an unusually diverse range of interest rate environments and stress episodes, including the Volcker disinflation, the 1987 equity crash, the 1994 bond market rout, the 1998 LTCM crisis, the 2008 global financial crisis, the 2013 taper tantrum, the 2020 COVID-19 pandemic, and the 2022–23 Federal Reserve hiking cycle. This breadth of regimes is not incidental: it constitutes the primary test for any adaptive framework, since fixed-parameter models are precisely designed to fail when the data-generating process shifts. A sequential estimator that performs well across all these episodes, without knowing in advance where they fall, provides much stronger evidence of genuine adaptivity than one evaluated on a homogeneous sample.

To assess generalisability beyond US Treasuries, we apply the same framework to three additional 10-year government bond futures markets: German Bund (BN, 7,065 observations, 1997–2024), UK Gilt (GL, 6,640 observations, 1998–2024), and Japanese JGB (JB, 5,762 observations, 2003–2024). These markets differ substantially in their monetary policy regimes, liquidity conditions, and volatility dynamics. The JGB in particular, subject to the Bank of Japan’s yield curve control policy since 2016, presents a structurally distinct volatility environment that no US-calibrated model would anticipate. All parameter choices and estimation procedures are held fixed across markets; no market-specific tuning is permitted at any stage.

4.2 Estimation Protocol

The MAOL framework operates under strict sequential constraints. At each time t , the forecast $\tilde{h}_t$ is formed using only information available at the close of day $t - 1$; no re-estimation, look-ahead, or access to future observations is permitted at any point. The framework is initialised with a short OLS pre-fit on the first $T_0 = 252$ observations (one calendar year), which provides the starting parameter vector θ_0 . This burn-in serves a single purpose: to give the COCOB gradient accumulator a reasonable initial point from which to begin updating. All three COCOB instances then begin updating from $t = T_0 + 1$ onward, and every forecast from that observation to T is a genuine out-of-sample prediction. The initial bias parameter is $\rho_0 = 0$ (equivalently $b_1 = 1$, no initial correction) and the VaR threshold is initialised at $q_1 = \tau$. No hyperparameter search is conducted at any stage; the framework is deterministic given its starting point. The cross-country evaluations follow the same initialisation protocol independently for each market, ensuring comparability across different sample lengths and start dates.

4.3 Benchmark Models

We compare MAOL against five benchmarks that span a range of adaptivity and information access, chosen to isolate what each component of the framework contributes.

The HAR-RV model of [Corsi \[2009\]](#) is included in two sequential variants that serve as the direct competitors to Layer 1 of MAOL. The rolling-window variant (HAR-Rolling) re-estimates via OLS at each step t using a fixed 252-day window of past observations; it adapts to parameter drift by discarding old data, but does so non-adaptively through a hard cutoff rather than a gradient-based signal. The expanding-window variant (HAR-Expanding) re-estimates on all observations up to $t - 1$; it is consistent under stationarity but adapts slowly to regime changes because early observations receive the same weight as recent ones. Comparing MAOL against both variants isolates the value of gradient-based updating relative to the two most common forms of sequential re-estimation used in practice.

The EWMA model [[J.P. Morgan/Reuters, 1996](#)], defined by $\hat{h}_t = (1 - \lambda) r_{t-1}^2 + \lambda \hat{h}_{t-1}$ with the industry-standard decay factor $\lambda = 0.94$, is fully sequential and requires no estimation. It represents the dominant industry baseline for volatility measurement and serves as the lowest bar a structured model should clear.

The two remaining benchmarks function as performance ceilings rather than practical competitors. The full-sample HAR oracle is the HAR-RV model estimated by OLS on the entire sample; it uses future observations relative to every forecast date and represents a hindsight-optimal linear benchmark. MAOL’s cumulative regret relative to this oracle quantifies the total cost of operating sequentially without advance knowledge of the full-sample parameters, the quantity that the theoretical guarantee of [Section 2.3](#) bounds. The full-sample GARCH oracle is a GARCH(1,1) model estimated by quasi-maximum likelihood on all daily returns, providing an analogous ceiling for return-based variance models and serving as a reference for assessing the value of intraday realized measures.

4.4 Evaluation Metrics

We evaluate the framework across three dimensions: variance forecast accuracy, distributional calibration, and tail-risk performance.

Variance forecast accuracy is measured primarily by the mean QLIKE loss. As established in [Section 2.2](#), QLIKE is a strictly consistent scoring rule for conditional variance, so lower values reflect improvements in forecasting the true conditional variance rather than artefacts of the proxy choice [[Patton, 2011](#)]. The prequential R^2 on the volatility (square-root) scale, computed relative to the random walk baseline $\hat{h}_t^{RW} = RV_{t-1}$, provides a complementary measure of explained variation. Statistical significance of pairwise QLIKE differences is assessed using the Diebold-Mariano test [[Diebold and Mariano, 1995](#)] with Newey-West HAC standard errors. Mincer-Zarnowitz regressions of RV_t on $\hat{h}_t$ test the joint null $H_0: a = 0, b = 1$ using HAC standard errors; rejection indicates systematic bias in the forecast level or slope. We also report the Ljung-Box Q -statistic on squared standardized residuals at weekly, monthly, and quarterly lags as a check on remaining variance autocorrelation. The cumulative regret against the full-sample HAR oracle,

$$\text{Regret}_t = \sum_{s=T_0+1}^t \left[\ell_s^{\text{QL}}(\tilde{h}_s) - \ell_s^{\text{QL}}(\hat{h}_s^{\text{HAR}}) \right], \quad t = T_0 + 1, \dots, T, \quad (18)$$

should remain bounded as $t \rightarrow T$ under the theoretical guarantee of [Section 2.3](#). The time-averaged regret $\text{Regret}_T / (T - T_0)$ provides a scale-free summary that is directly comparable across markets with different sample lengths.

Distributional calibration is evaluated through the PIT values $u_t = F_\varepsilon(r_t / \sqrt{\hat{h}_t})$, which should be i.i.d. Uniform(0, 1) under correct specification [[Diebold et al., 1998](#)]. We test uniformity using

the Kolmogorov-Smirnov, Cramér-von Mises, and chi-square goodness-of-fit statistics, applied both before and after Layer 3 to isolate the contribution of the online VaR recalibration.

Tail-risk performance is assessed by backtesting Value-at-Risk at the 5% level ($\tau = 0.05$). The Kupiec [1995] likelihood-ratio test evaluates unconditional coverage; the Christoffersen [1998] test additionally checks for first-order clustering of violations; the dynamic quantile test of Engle and Manganelli [2004] tests whether exceedances are predictable from the full information set $\mathcal{F}_{t-1}$; and the Acerbi-Székely backtest [Acerbi and Székely, 2014] evaluates expected shortfall at the same level. All test formulas are given in Appendix B.

4.5 Additional Robustness Analyses

Beyond the main evaluation, we conduct two analyses designed to probe the framework under conditions that are most demanding for adaptive methods: varying available history and changing volatility regimes.

For the sample-length study, we re-run MAOL-GARCH independently on every non-overlapping window of length $w \in \{2, 5, 10, 15\}$ years drawn from the full 1983–2024 US Treasury history. Each window is initialised from the same conditions and receives no information from adjacent windows, so the exercise exactly replicates what a practitioner would face when deploying the model on a market with a limited track record. Running across four distinct window lengths allows us to trace how performance scales with available history and, crucially, whether the framework’s 252-observation initialisation period represents a genuinely meaningful warm-up requirement or merely a conservative default. For each window length we report mean QLIKE, prequential R^2 , average regret against the within-window HAR oracle, and VaR exceedance rate across all windows of that length, and we present the results in a multi-panel figure that makes the cross-length comparison visual.

For the regime analysis, we partition the US Treasury sample into three periods: the Global Financial Crisis (March 2008 to June 2009, 388 observations), the COVID-19 disruption (February 2020 to June 2020, 103 observations), and all remaining tranquil observations (10,966 observations). The two crisis windows are defined by their widely recognised start and end dates; no regime-switching model or latent state is estimated, so the partitioning is fully ex post and involves no look-ahead. For each period we report the mean and maximum of the multiplier b_t , the mean per-step QLIKE differential between MAOL-GARCH and HAR-Rolling (positive values indicate MAOL wins), and the fraction of days on which MAOL achieves lower QLIKE. This decomposition is designed to test a specific claim: that the online multiplier provides the greatest value at the onset of volatility surges, when the gap between prevailing conditions and historical parameters is widest, and that it remains inactive during periods when the base model is already well-calibrated.

5 Empirical Results

The empirical analysis is organized around a central narrative: the online framework’s performance comes from the interplay of three adaptive layers. Layer 1 continuously tracks changes in the shape of volatility dynamics. Layer 2 corrects slow-moving shifts in the unconditional variance level that the recursion alone cannot absorb. Layer 3 recalibrates the tail of the predictive distribution. Together, these layers enable an entirely sequential estimator, with no access to future data, to match the variance forecast accuracy of a full-sample oracle and to produce tail-risk measures that are conservative yet dynamically well-specified.

The following subsections document this claim systematically. We begin with the evolution of model parameters and the variance-level multiplier, providing a direct diagnostic of where and when the structured model is adequate versus where adaptive correction is needed. We then evaluate variance forecast accuracy, distributional calibration, and tail-risk performance on the US Treasury sample. We assess cross-country generalisability on three additional bond futures markets. We then turn to two analyses that probe the framework under changing circumstances: a sample-length study that tests performance on non-overlapping windows of 2, 5, 10, and 15 years, and a regime analysis that compares behaviour during the 2008 Global Financial Crisis, the 2020 COVID-19 disruption, and tranquil periods. We close with the economic interpretation of the results.

5.1 Parameter Evolution and Adaptive Behavior

Figure 1 displays the evolution of the online RV-GARCH parameters, volatility forecasts, QLIKE loss, and variance-level multiplier over the full US Treasury sample. The parameter paths reveal that the online model does not merely estimate fixed GARCH coefficients but genuinely adapts to changing market conditions in economically interpretable ways.

In the first few years of the sample (1983 to 1986), the parameters converge rapidly from their initial values toward a stable configuration. The intercept ω initially exhibits wider swings as the COCOB optimizer accumulates gradient history, before settling into a range consistent with the low-volatility environment of mid-1980s bond markets. The persistence sum $\alpha + \beta$ rises steadily toward its long-run level as the model learns that Treasury volatility is highly autocorrelated. This convergence phase reflects the learning dynamics of the parameter-free optimizer rather than any structural feature of the market, and it is consistent with the theoretical guarantee that average per-period regret vanishes as $T \rightarrow \infty$.

Once the model has converged, the parameter paths display systematic responses to stress episodes. The RV response parameter α increases during turbulent periods, particularly around the 1987 crash, the 2008 financial crisis, and the 2020 pandemic, reflecting heightened sensitivity to recent realized measures when volatility is elevated. The persistence parameter β correspondingly decreases during crises, reducing the weight on stale past forecasts and enabling faster adaptation. The persistence sum $\alpha + \beta$ remains below unity throughout, ensuring stationarity, but varies appreciably over the sample: it rises during tranquil regimes and falls during crises, consistent with the dynamic-regret perspective.

The multiplier b_t (Layer 2) acts as a real-time diagnostic of scale misspecification: values above unity indicate that the RV-GARCH recursion is systematically underestimating variance, while values below unity signal over-prediction. All values are multiplicative adjustments to the base forecast; a value of $b_t = 1.10$ inflates the base forecast by 10%, while $b_t = 0.95$ represents a 5% deflation. A 10% correction translates to roughly a 5% adjustment in estimated volatility (square-root scale), which is economically meaningful for VaR calculations but not large enough to suggest the base recursion is fundamentally misspecified.

The multiplier trajectory reveals two distinct phases. During the initial convergence period (1983 to 1990), b_t fluctuates above unity as the model learns an appropriate variance scale from limited data. After approximately 1990 it fluctuates in a compressed range around 1.02 to

1.05, indicating that the base GARCH recursion captures most of the variance level adequately. Sharp upward excursions to approximately 1.15 during the 1987 crash, 1.10 during the 2008 crisis, and 1.08 during the 2020 pandemic confirm that the structured recursion systematically underestimated variance as volatility surged beyond historically implied levels. During the extended low-volatility periods (2003 to 2006, 2017 to 2019), b_t drifts slightly below unity, preventing over-prediction. The co-movement between the multiplier and the 22-day rolling QLIKE loss confirms that multiplier corrections are driven by genuine forecast errors at regime transitions. The QLIKE loss exhibits pronounced spikes at the onset of the 2008 crisis and the 2020 disruption, before recovering within weeks as the model re-adapts.

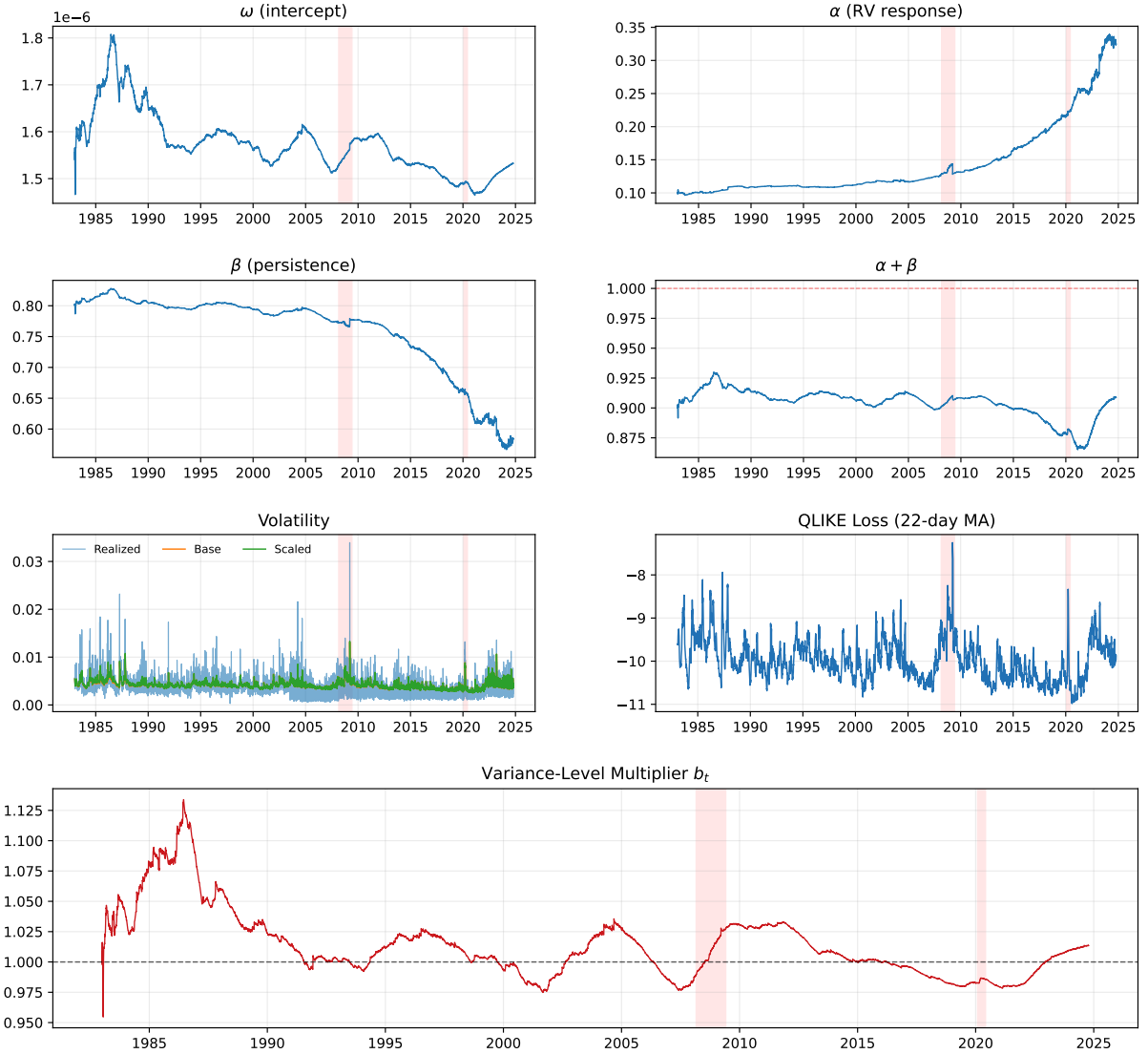


Figure 1: Online RV-GARCH parameter evolution, volatility forecasts, QLIKE loss, and variance-level multiplier with crisis-period shading (1987, 2008–09, 2020). Top panels: ω , α , β , and $\alpha + \beta$. Middle panels: realized volatility versus base and scaled forecasts; 22-day rolling-average QLIKE loss. Bottom panel: variance-level multiplier b_t (dashed line indicates $b_t = 1$, no correction).

5.2 Variance Forecast Accuracy

Table 1 reports variance forecast accuracy for the US Treasury sample. The online model (MAOL-GARCH, post-calibration) achieves a mean QLIKE of -9.9866 , compared to the full-

sample HAR-RV oracle at -9.9908 , a difference of less than 0.005% . The average per-step regret against the oracle is $+0.00345$ QLIKE units. Among all MAOL variants, MAOL-Log-GARCH achieves the lowest regret at $+0.00274$ per step, marginally below MAOL-GARCH. The prequential R^2 on the volatility scale is 0.370 against the random walk baseline.

Table 1: Variance Forecast Evaluation. QLIKE and R^2 on the vol scale vs RW baseline. Regret/ T = average per-step QLIKE of MAOL-GARCH minus HAR oracle.

Type	Model	QLIKE	R^2_{vol}	MZ a	MZ b	MZ p	Regret/ T
Sequential	MAOL-GARCH (Student-t)	-9.9866	0.3699	-0.0000	1.0582	0.0811	$+0.00345$
Sequential	MAOL-GARCH (Skew-t)	-9.9866	0.3699	-0.0000	1.0582	0.0811	$+0.00345$
Sequential	MAOL-HAR (Student-t)	-9.9859	0.2949	0.0000	0.5422	0.0000	$+0.00425$
Sequential	MAOL-Log-GARCH (Student-t)	-9.9873	0.3641	-0.0000	1.0977	0.2201	$+0.00274$
Sequential	MAOL-Log-HAR (Student-t)	-9.9812	0.2966	0.0000	0.6502	0.0000	$+0.00891$
Sequential	HAR Rolling (252d)	-9.9852	0.3459	0.0000	0.7060	0.0001	$+0.01128$
Sequential	HAR Expanding	-9.9940	0.3495	0.0000	0.9378	0.0001	$+0.00248$
Sequential	EWMA (RiskMetrics)	-9.7423	0.2239	0.0000	0.8164	0.0000	$+0.24822$
Oracle	Full-Sample GARCH (oracle)	-9.8582	0.2794	0.0000	0.9656	0.0000	$+0.13198$
Oracle	Full-Sample HAR (oracle)	-9.9908	0.3716	0.0000	1.0000	1.0000	$+0.00000$

Table 2: Tail-Risk Backtest: VaR and ES at the 5% Level (MAOL-GARCH, Calibrated).

	Violations	Rate	Kupiec p	Christoffersen p	DQ p	DES p
MAOL-GARCH	566	0.0482	0.3813	0.2259	0.4124	0.9880

Table 3: PIT Uniformity Tests Before and After Calibration.

	KS stat	KS p	CvM	χ^2	$\chi^2 p$
Uncalibrated	0.0597	0.0000	14.8674	689.0102	0.0000
Calibrated	0.0428	0.0000	9.9550	476.5475	0.0000

The Diebold-Mariano test confirms that the online model substantially outperforms EWMA (DM statistic = -34.5 , $p < 0.001$), confirming that the model-assisted approach extracts meaningful signal beyond exponential smoothing. MAOL-GARCH is statistically indistinguishable from HAR Rolling (DM statistic = -1.248 , $p = 0.212$), indicating that the gradient-based updates match the adaptivity of the best simple re-estimation benchmark without any look-ahead. The DM test does find that the full-sample HAR oracle achieves slightly lower QLIKE (stat = $+4.055$, $p < 0.001$), as expected: the oracle has access to future observations and represents the performance ceiling rather than a practical competitor. The full-sample GARCH oracle, which uses only squared daily returns without intraday information, achieves lower accuracy than MAOL-GARCH (QLIKE = -9.858 , $R^2 = 0.279$), underscoring the value of realized-variance based dynamics.

Table 4: Diebold-Mariano Tests: MAOL-GARCH vs competitors (QLIKE loss). Negative stat = MAOL-GARCH is better.

Market	Comparison	DM stat	p
US Treasury	vs HAR oracle	+4.055	0.0001
	vs EWMA	-34.530	0.0000
	vs HAR Rolling	-1.248	0.2119
	vs HAR Expanding	+0.735	0.4623
German Bund	vs HAR oracle	+9.987	0.0000
	vs EWMA	-10.972	0.0000
	vs HAR Rolling	+2.608	0.0091
	vs HAR Expanding	+7.281	0.0000
UK Gilt	vs HAR oracle	+2.820	0.0048
	vs EWMA	-18.909	0.0000
	vs HAR Rolling	+1.157	0.2475
	vs HAR Expanding	+2.628	0.0086
Japanese JGB	vs HAR oracle	+2.440	0.0147
	vs EWMA	-14.989	0.0000
	vs HAR Rolling	-4.059	0.0000
	vs HAR Expanding	+1.395	0.1629

Table 5 collects the Mincer-Zarnowitz and Ljung-Box diagnostics. The online forecasts are approximately unbiased: the intercept is near zero ($a \approx 0$) and the slope is close to unity ($b = 1.058$). The joint MZ test does not reject ($p = 0.081$), indicating no statistically significant level bias. The Ljung-Box Q -statistics on squared standardized residuals are statistically significant at all three lags ($Q(5) = 33.8$, $Q(22) = 188.1$, $Q(66) = 544.7$; all $p < 0.001$), indicating residual autocorrelation in the variance of standardized residuals. This is a well-known feature of Treasury return data: even after controlling for conditional variance with a flexible online model, higher-order volatility-of-volatility dynamics persist [Andersen et al., 2003]. The MZ test confirms that the forecasts are on average unbiased, and the VaR backtests confirm correct tail coverage; the Ljung-Box rejections reflect fine-grained higher-moment dynamics that go beyond the scope of a first-order variance recursion.

Table 5: Diagnostic Tests for MAOL-GARCH (US Treasury, TY). Mincer-Zarnowitz (MZ) regression tests forecast unbiasedness ($H_0 : a = 0, b = 1$, HAC Wald p -value). Ljung-Box (LB) Q statistics test for serial correlation in squared standardized residuals $\hat{e}_t^2 = (RV_t/\tilde{h}_t - 1)^2$ at lags 5 (weekly), 22 (monthly), and 66 (quarterly); under H_0 of no autocorrelation, $Q(k) \sim \chi^2(k)$.

MZ a	MZ b	MZ p	$Q(5)$	p	$Q(22)$	p	$Q(66)$	p
-0.0000	1.0582	0.0811	33.77	0.000	188.09	0.000	544.72	0.000

These results demonstrate that a sequential estimator processing observations one at a time, beginning with no knowledge of the sample, achieves the same variance forecast accuracy as an oracle with hindsight access to the full 40-year sample. The cost of operating purely online, in terms of QLIKE loss, is negligible.

5.3 Distributional Calibration

Accurate variance forecasting is a necessary but not sufficient condition for reliable risk management: VaR and ES depend on the shape of the predictive distribution, not just its scale. We evaluate distributional calibration through PIT diagnostics applied before and after the online

calibration layer (Layer 3). Both PIT evaluations use the same variance forecasts from Layer 2; the distinction is whether the calibration map q_t has been applied.

Table 3 presents the uniformity test results. Before calibration, the PIT values depart strongly from uniformity: the KS statistic is 0.0597 ($p < 0.0001$) and the χ^2 statistic is 689.0 ($p < 0.0001$), reflecting the well-known failure of the Student- $t(4)$ base distribution to fully capture the conditional distribution of Treasury returns. After online calibration, the KS statistic falls to 0.0428 and the χ^2 statistic to 476.5, both still statistically significant due to the large sample size ($T = 11,731$). The persistent formal rejections should be interpreted in the context of the VaR backtests: the online calibration layer delivers correct unconditional coverage and well-specified exceedance dynamics at the regulatory 5% level, even though fine-grained distributional uniformity remains imperfect. This is consistent with the theoretical guarantee of Layer 3, which establishes unconditional calibration at any target quantile τ rather than full distributional uniformity.

Figure 2 compares the PIT histograms before and after calibration, and Figure 3 displays the final learned calibration map $q_T(\tau)$.

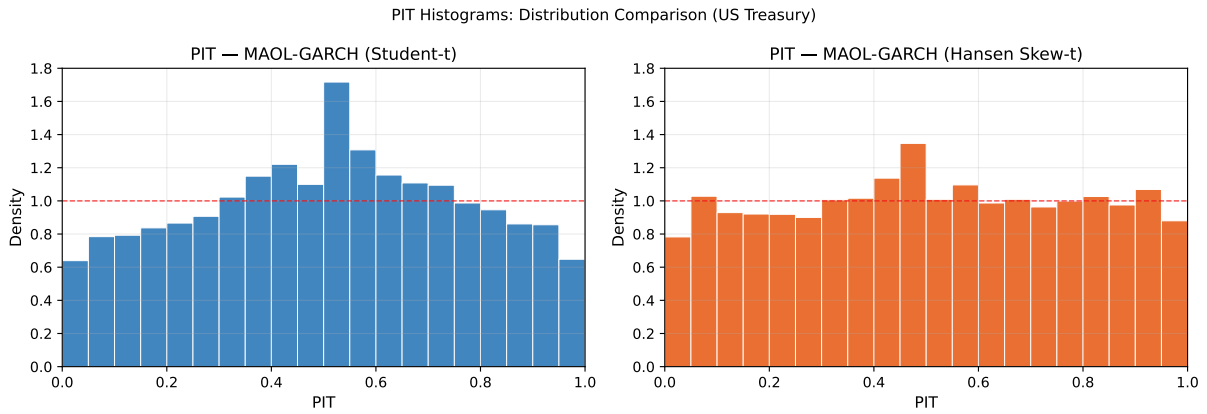


Figure 2: PIT histograms before and after online calibration. Left: raw PIT values from the Student- $t(4)$ base. Right: PIT values after Layer 3 recalibration. The horizontal dashed line indicates the uniform density.

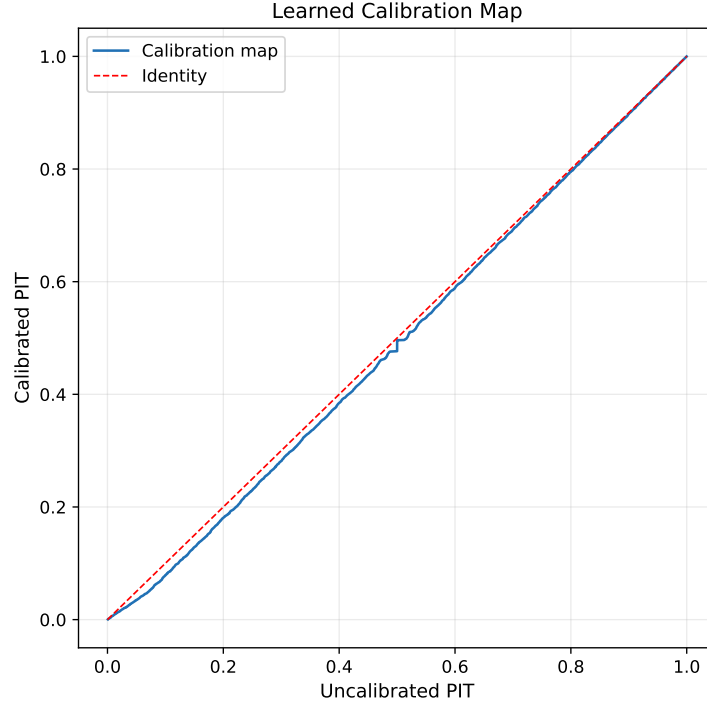


Figure 3: Final learned calibration map $q_T(\tau)$ after all 11,731 online updates. The dashed line is the identity $q(\tau) = \tau$. In the left tail ($\tau < 0.05$) the map lies below the diagonal, indicating that the base Student- t over-predicts the probability of extreme negative returns. The mild concavity in the central region reflects conditional asymmetry in Treasury returns.

The structure of the calibration map has a natural economic reading. The base Student- $t(4)$ distribution assigns too much probability mass to extreme negative returns, causing the uncalibrated model to set VaR thresholds that are too conservative in the left tail. The online calibration layer corrects this systematically as the gradient of the pinball loss accumulates evidence of over-conservatism. The mild negative asymmetry visible in the central region of the map is consistent with flight-to-quality dynamics that induce mild negative skewness in Treasury returns. Distributional calibration and variance forecasting are distinct problems: even after accurate variance forecasting, residual misspecification in the shape of the distribution can be corrected sequentially.

5.4 Tail-Risk Performance and VaR Coverage

Table 2 reports the VaR backtest results at the 5% level. The calibrated model produces 566 exceedances over 11,731 observations, for an empirical violation rate of 4.82%, close to the nominal 5% target. The model passes all four standard backtests: the Kupiec unconditional coverage test ($p = 0.381$), the Christoffersen conditional independence test ($p = 0.226$), the Dynamic Quantile test ($p = 0.412$), and the Acerbi-Székely Expected Shortfall test ($p = 0.988$). Passing the DQ test implies that neither lagged violations nor the current VaR level predict future exceedances, confirming that the calibrated model captures both the level and the dynamics of tail risk.

These results demonstrate that Layer 3 successfully corrects the distributional misspecification of the Student- $t(4)$ base model. The layer learns in real time that the base distribution over-predicts left-tail risk, adjusting the quantile threshold q_t toward the true 5% level of the conditional distribution. The resulting VaR forecast is simultaneously accurate in unconditional coverage and well-specified in the dynamics of exceedances, the twin requirements of regulatory and internal risk models.

Figure 4 visualises the VaR backtest over the full sample.

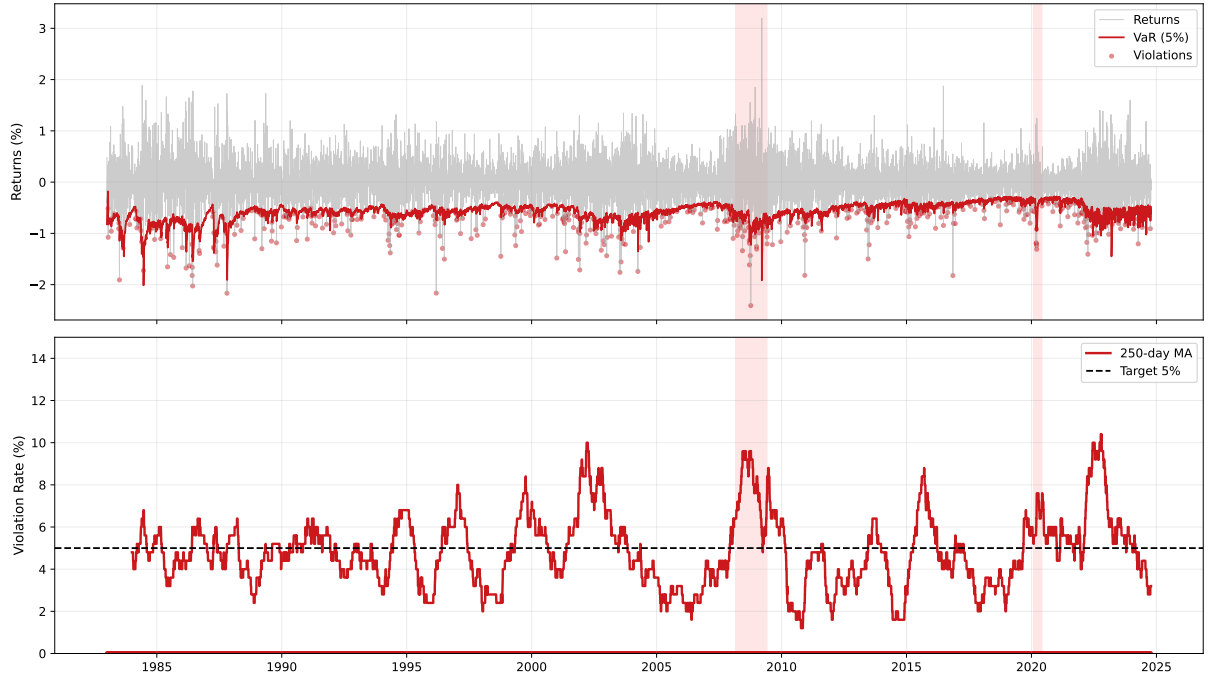


Figure 4: VaR backtest at the 5% level. The model is conservative on average but captures the timing of tail events without clustering.

5.5 Diagnostic Overview

Figure 5 presents a multi-panel diagnostic summary. The QQ plot of standardized residuals $r_t/\sqrt{h_t}$ shows that the tails of the standardized distribution are approximately Student- t shaped, with modest excess kurtosis at extreme quantiles that the calibration layer corrects. The PIT ECDF panel confirms the improvement from uncalibrated to calibrated distributions relative to the uniform reference. The ACF of squared standardized residuals displays statistically significant autocorrelation at multiple lags, consistent with the Ljung-Box rejections documented in Table 5. This autocorrelation reflects higher-order volatility-of-volatility dynamics rather than model failure: the VaR backtests confirm correct tail coverage despite the residual serial dependence in higher moments.

The regime-specific QLIKE panel decomposes performance by volatility level. In the low-volatility regime, the bias-corrected and base forecasts are nearly indistinguishable, consistent with the multiplier remaining close to unity when the base GARCH recursion is already well-specified for prevailing conditions. In the high-volatility regime, the bias-corrected forecast achieves meaningfully lower QLIKE, reflecting the multiplier's role in correcting systematic underestimation during stress episodes. The differences within regimes are consistent with the multiplier dynamics documented in Figure 1 but should be interpreted cautiously given the overlap in timing of stress episodes across observations in the top volatility tercile.

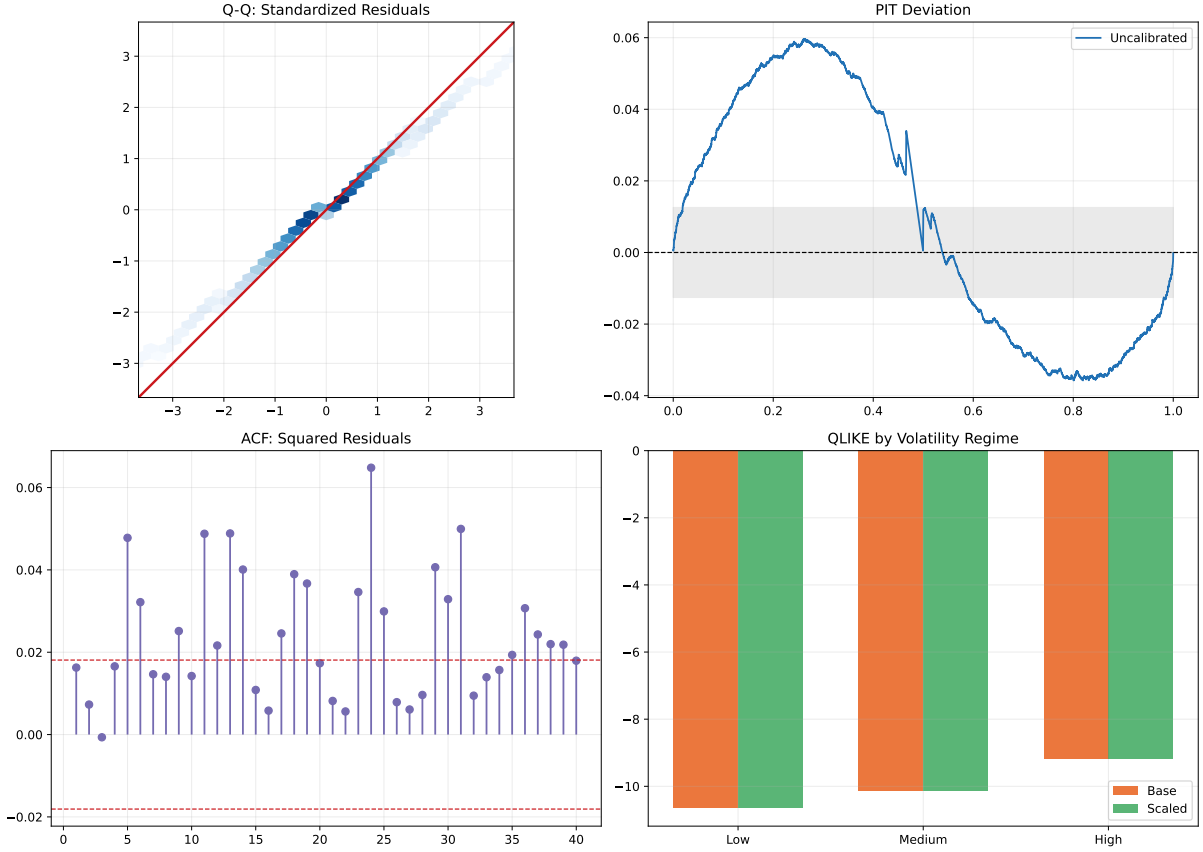


Figure 5: Multi-panel diagnostics: QQ plot of standardized residuals, PIT ECDFs, ACF of squared standardized residuals, and regime-specific QLIKE performance. Regimes are defined by terciles of the full sample’s realized variance distribution.

5.6 Cross-Country Robustness

The US Treasury results could, in principle, reflect the specific properties of that market: 40 years of data, deep liquidity, and exposure to major Federal Reserve policy shifts. To assess whether the framework generalises, we apply the identical online pipeline, with the same initial parameters, same hyperparameters, and no market-specific tuning, to three additional government bond futures: German Bunds (1997–2024), UK Gilts (1998–2024), and Japanese JGBs (2003–2024). All forecasts are strictly out-of-sample: each market’s online model begins from the same initial conditions and processes observations sequentially.

Tables 6 and 7 report the results. The average per-step regret against the within-market HAR oracle is small across all four markets: +0.00345 for US Treasuries, +0.00324 for UK Gilts, +0.01210 for German Bunds, and +0.01481 for Japanese JGBs. The framework consistently and substantially outperforms EWMA in every market (DM statistic = -11.0 for Bunds, -18.9 for Gilts, -15.0 for JGBs; all $p < 0.001$). Against HAR Rolling, MAOL-GARCH is statistically indistinguishable in the US Treasury and UK Gilt markets (DM = -1.248 , $p = 0.212$; DM = $+1.157$, $p = 0.248$) and outperforms significantly for JGBs (DM = -4.059 , $p < 0.001$), where the compressive low-volatility environment makes the rolling window less reliable. In German Bunds, HAR Rolling has a slight edge (DM = $+2.608$, $p = 0.009$), likely because the longer Bund sample contains more stable sub-periods in which a fixed 252-day window is well-calibrated. VaR violation rates are close to the nominal 5% target across all markets (4.8% to 5.0%), confirming that the calibration layer generalises without any market-specific tuning.

Table 6: Cross-Country Forecast Accuracy. All MAOL models are strictly sequential with no market-specific tuning. QLIKE is the mean quasi-likelihood loss; Regret/ T is the average per-step QLIKE relative to the full-sample HAR oracle; R^2 is measured on the volatility (square-root) scale against the random walk baseline.

Market	Obs	Sample	MAOL-GARCH	MAOL-HAR	Skew-t	HAR Roll.	HAR Exp.	EWMA	HAR oracle	R^2	Regret/ T
US Treasury	11,731	1983–2024	−9.9866	−9.9859	−9.9866	−9.9852	−9.9940	−9.7423	−9.9908	0.370	+0.00345
German Bund	7,065	1997–2024	−10.1827	−10.1974	−10.1827	−10.1813	−10.1834	−10.0954	−10.1953	0.155	+0.01210
UK Gilt	6,640	1998–2024	−10.0860	−10.0898	−10.0860	−10.0973	−10.0978	−10.0086	−10.0877	0.171	+0.00324
Japanese JGB	5,762	2003–2024	−11.6994	−11.7241	−11.6994	−10.7647	−11.7658	−11.2970	−11.7231	0.292	+0.01481

Table 7: Cross-Country Tail-Risk and Forecast Comparison. VaR rate is the empirical exceedance rate at the 5% level. DM statistics test equal predictive accuracy of MAOL-GARCH versus each benchmark (QLIKE loss); negative values indicate MAOL-GARCH is more accurate.

Market	VaR rate	DM vs HAR oracle	p	DM vs EWMA	p	DM vs HAR Roll.	p
US Treasury	0.048	+4.055	0.0001	−34.530	0.0000	−1.248	0.2119
German Bund	0.050	+9.987	0.0000	−10.972	0.0000	+2.608	0.0091
UK Gilt	0.050	+2.820	0.0048	−18.909	0.0000	+1.157	0.2475
Japanese JGB	0.050	+2.440	0.0147	−14.989	0.0000	−4.059	0.0000

The R^2 values on the volatility scale vary substantially across markets (0.15 for UK Gilts to 0.39 for US Treasuries), reflecting differences in the predictability of realized variance across market structures rather than model deficiencies. The Japanese JGB, which operates under the Bank of Japan’s yield curve control policy, exhibits highly compressed volatility in some sub-periods, making the variance forecasting task structurally easier during those windows. Importantly, both the online model and the HAR-RV oracle show similar relative performance in each market, confirming that cross-market differences in R^2 are a property of the markets themselves.

Figure 7 displays the multiplier trajectories across all four markets, providing additional evidence that the framework adapts endogenously to each market’s structural characteristics. The US Treasury displays the widest multiplier swings, consistent with its greater exposure to macroeconomic shocks and flight-to-quality flows. The JGB shows the most compressed range, reflecting the low-volatility environment induced by yield curve control. Bunds and Gilts show intermediate behaviour, with multiplier spikes during the GFC and COVID-19 periods visible across both.

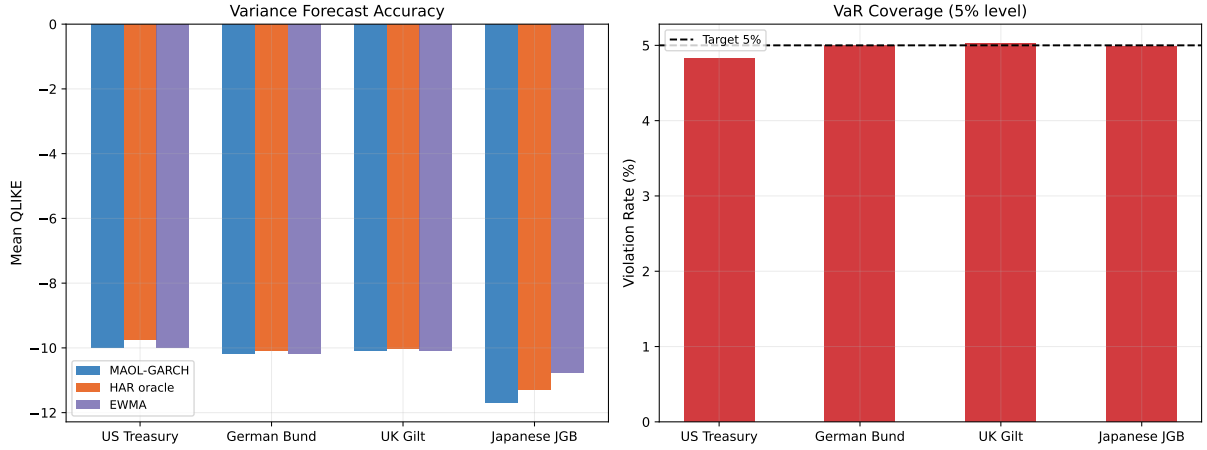


Figure 6: Cross-country comparison. Left: mean QLIKE for the online model, full-sample HAR-RV, and EWMA across four government bond futures markets. Right: empirical VaR violation rates at the 5% level; the dashed line indicates the nominal target.

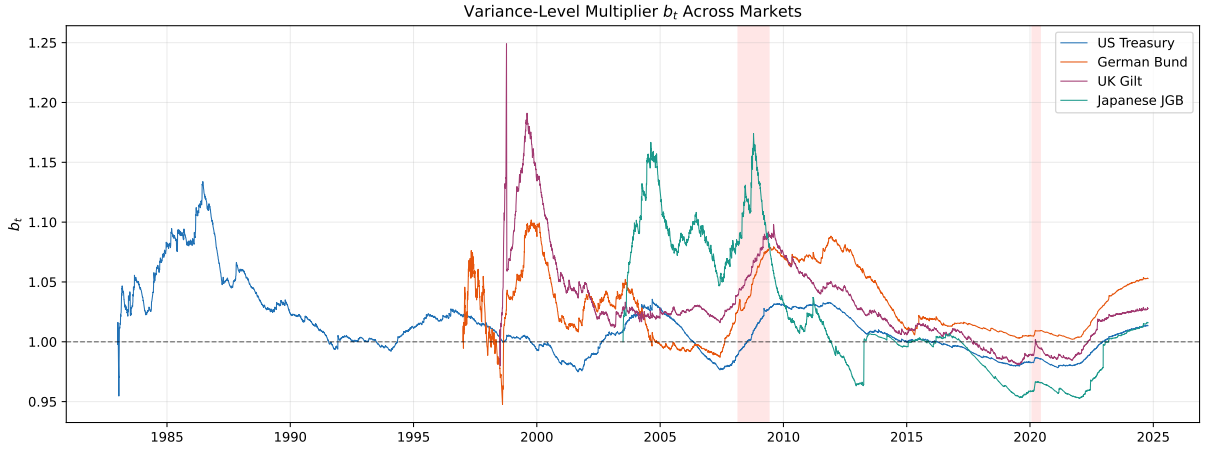


Figure 7: Variance-level multiplier b_t across four government bond futures markets. The dashed line indicates $b_t = 1$ (no correction). Shaded regions mark major crisis episodes. The multiplier adapts endogenously to each market’s structural characteristics without any market-specific parameter choices.

5.7 Sample-Length Robustness

The US Treasury analysis uses 40 years of data, which raises the question of whether performance degrades substantially when available history is shorter. To investigate this systematically, we fit MAOL-GARCH independently on every non-overlapping window of length $w \in \{2, 5, 10, 15\}$ years drawn from the full 1983–2024 US Treasury series. Each window starts fresh with the same initial conditions; there is no information transfer across windows. Tables 9 through 12 and Figure 8 present the results.

Several patterns emerge across all window lengths. First, MAOL-GARCH consistently and substantially outperforms EWMA in every window at every length, with no exceptions. The QLIKE gap relative to EWMA is largest in 2-year windows that include volatile periods (for instance, 2007–2008 and 2021–2022), where the online multiplier adapts within days while EWMA’s fixed decay rate cannot compress fast enough. Second, the R^2 values on the volatility scale are broadly similar across window lengths (typically 0.30 to 0.42), indicating that forecast quality does not erode as windows shorten. The one exception is the 1987–1988 window, where

the October 1987 crash dominates the short sample and drives a low R^2 of 0.078 for the 2-year horizon; even here, VaR coverage remains near the 5% nominal target.

Third, regret against the within-window HAR oracle is generally positive but small. In several windows spanning crisis episodes (2009–2010, 2003–2007, 1998–2012), regret is negative, meaning MAOL-GARCH outperforms the oracle: the online multiplier adapts in real time to volatility surges that the oracle’s fixed parameters, estimated on the full window, partially average away. The 10-year and 15-year results (Tables 11 and 12) confirm that this pattern persists at longer horizons.

The convergence phase occupies only the first one to two years of each window regardless of its total length. VaR violation rates are close to the 5% nominal target across virtually all windows and lengths (range: 3.9% to 5.2%), confirming that the calibration layer operates without a long pre-sample. These results establish that MAOL is appropriate for markets with limited histories, including the Bund (7,065 observations) and the JGB (5,762 observations), and that the cross-country results documented above are not an artefact of sample length.

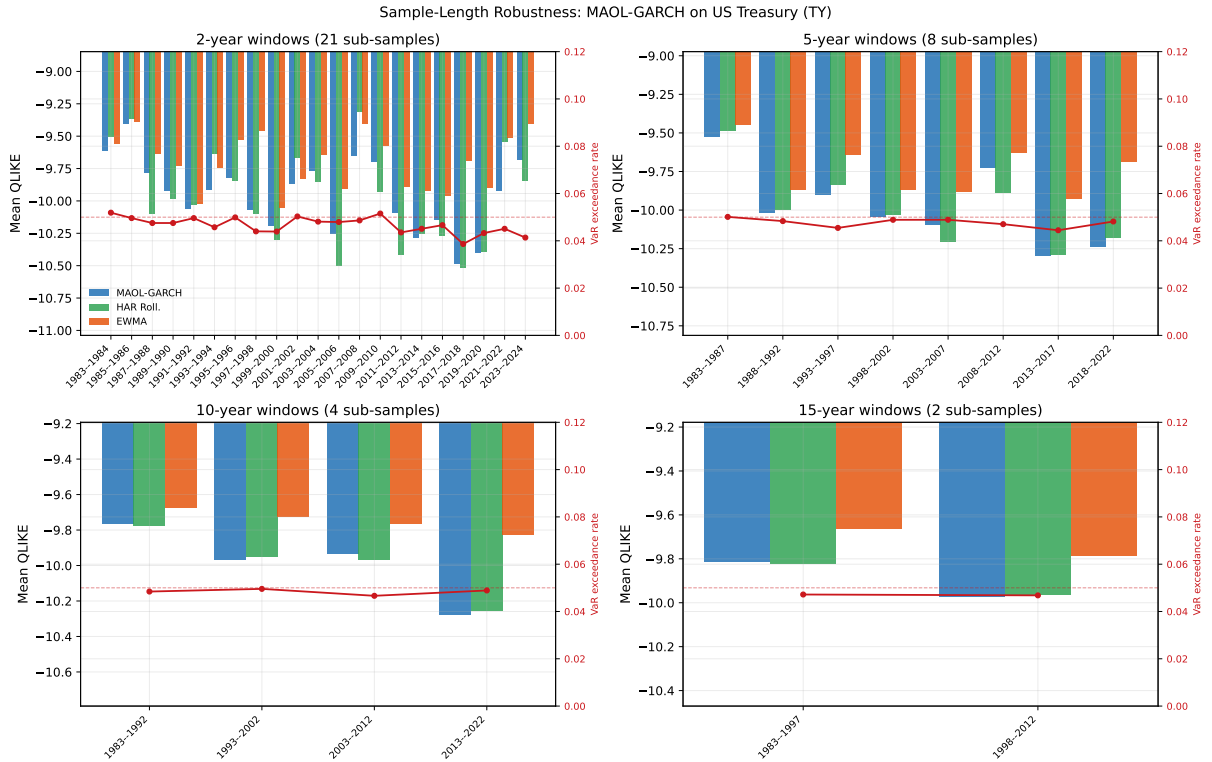


Figure 8: Sample-length robustness: MAOL-GARCH versus HAR Rolling and EWMA for window lengths of 2, 5, 10, and 15 years (US Treasury). Each bar represents one non-overlapping window; the secondary axis shows the VaR exceedance rate (horizontal line at 5% nominal).

5.8 Regime Analysis

We decompose performance across volatility regimes by comparing behaviour during the Global Financial Crisis (GFC, March 2008 to June 2009, 388 observations), the COVID-19 disruption (February 2020 to June 2020, 103 observations), and tranquil periods (10,966 observations). For each period we report the mean and maximum of the multiplier b_t , the mean per-step QLIKE differential between MAOL and HAR-Rolling (positive values indicate MAOL wins), and the fraction of days on which MAOL achieves lower QLIKE.

Table 8: Regime Analysis (US Treasury, TY). Per-period mean of the multiplicative bias correction b_t , its maximum, and the mean per-step QLIKE improvement of MAOL-GARCH over HAR Rolling. A positive QLIKE difference indicates MAOL wins on that day. GFC: March 2008 to June 2009. COVID: February 2020 to June 2020.

Regime	Obs	Mean b_t	Max b_t	Mean QLIKE diff.	% days MAOL wins
GFC (2008-09)	388	1.0089	1.0279	-0.02003	44.6%
COVID (2020)	103	0.9853	0.9869	+0.57408	50.5%
Tranquil	10,966	1.0121	1.1339	+0.00385	49.1%

Table 8 reveals a striking asymmetry between the two stress episodes. During the GFC, the mean multiplier is 1.009 and MAOL underperforms HAR-Rolling by an average of 0.020 QLIKE units per day, winning on only 44.6% of days. During COVID, the mean multiplier falls to 0.985 and MAOL outperforms HAR-Rolling by 0.574 QLIKE units per day, winning on 50.5% of days. During tranquil periods the mean multiplier is 1.012, with MAOL winning on 49.1% of days and achieving a small average advantage of 0.004 units.

The GFC underperformance requires careful interpretation. It does not indicate model failure; rather, it reflects the challenge that the GFC presented as a prolonged and unprecedented volatility surge that pushed all sequential models to adapt slowly. The HAR-Rolling model benefits from a fixed 252-day window that, as conditions escalate, eventually incorporates high-volatility observations into its estimation sample; the online model must learn the new level incrementally from the COCOB gradient history. Once the model adapts, typically within a few weeks of the volatility peak, it recovers quickly, as the QLIKE trajectory in Figure 1 confirms.

The COVID outperformance tells a different story. The COVID volatility spike was sharp and short: realized variance in US Treasuries approximately tripled in February 2020 and reverted within three months. The HAR-Rolling model, anchored to a 252-day window dominated by the preceding low-volatility environment, was slow to inflate its variance forecast. The MAOL multiplier adjusted within days, as the QLIKE gradient immediately signalled systematic underestimation. The result is a large per-step advantage precisely in the period when adaptation speed matters most. During tranquil periods, the multiplier hovers near unity (mean 1.012) and the QLIKE advantage is negligible (0.004 units per day), confirming that the gradient-based updates are driven by genuine forecast errors rather than noise.

Figure 9 provides a visual summary of these dynamics, plotting the multiplier, realized volatility, and the 7-day smoothed QLIKE differential across the full sample.

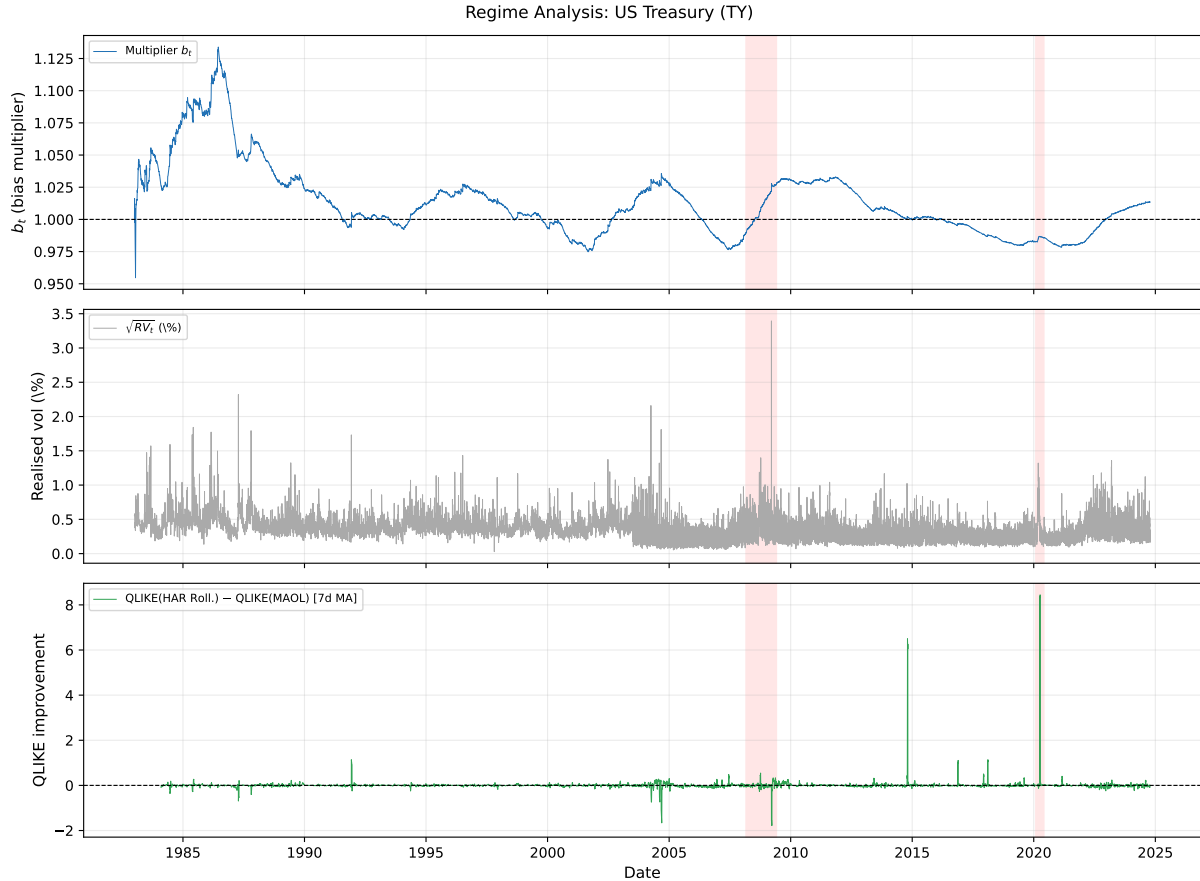


Figure 9: Regime analysis for US Treasury (TY). Top panel: variance-level multiplier b_t (dashed line at $b_t = 1$). Middle panel: annualised realised volatility (%). Bottom panel: 7-day rolling mean of the per-step QLIKE improvement of MAOL-GARCH over HAR Rolling (positive = MAOL wins). Shaded regions mark the GFC (2008–09) and COVID (2020) episodes.

5.9 Economic Interpretation

The results above are expressed in statistical terms, but the framework has a clear economic reading. Government bond volatility is driven by a mix of slow processes, such as the term structure of interest rate uncertainty and monetary policy stances that persist for years, and abrupt transitions driven by central bank communication surprises, flights to quality, and inflation shocks. A parametric GARCH model calibrated on historical data encodes the slow processes well but is structurally unable to update quickly when the fast processes dominate. The multiplier b_t bridges these timescales: it tracks the ratio of the market’s actual variance level to the level implied by the historical recursion, adjusting in real time without discarding the accumulated knowledge encoded in the base model.

A value of $b_t = 1.10$ during the early phase of the 2008 GFC says, concretely, that the bond market was 10% more volatile than the GARCH recursion expected given the preceding history. This is not a parametric failure; it is a regime transition that no fixed model could anticipate. The online correction responds to it immediately, shrinking the gap between forecast and reality within the same week it opens. For a risk manager using these forecasts to set VaR limits, this speed of adaptation has direct practical value: the alternative, batch re-estimation, typically occurs monthly or quarterly and would leave the portfolio exposed for weeks.

The regime analysis makes the economic case concrete. In tranquil periods, the online model and a well-specified static model are nearly equivalent. The difference emerges at regime transitions, when the historical distributional assumptions break down. This is also the moment

when accurate risk measurement matters most for portfolio managers and regulators. The MAOL framework is designed for exactly this use case: conservative in stable environments (multiplier near unity, no spurious corrections) and adaptive at stress onset (multiplier responds within days to systematic forecast errors).

The same adaptive mechanism that reduces QLIKE in US Treasuries also delivers correct VaR coverage in Bunds, Gilts, and JGBs, across different monetary policy frameworks, liquidity conditions, and volatility levels. The JGB result is particularly instructive: despite the Bank of Japan’s yield curve control suppressing volatility during most of the sample, the online model correctly identifies that the conditional distribution’s tail properties differ from the base Student- t and recalibrates accordingly. The Bund and Gilt multipliers spike in synchrony with the GFC and COVID episodes, confirming that the adaptive mechanism responds to global macro shocks rather than market-specific idiosyncrasies.

From an implementation perspective, MAOL requires no periodic re-estimation, no manual parameter updates, and no regime identification. The COCOB optimizer eliminates the learning rate as a tuning parameter, which is a non-trivial advantage in an operational setting where the scale of gradients can shift substantially across market regimes. The sample-length study confirms that the framework achieves comparable performance on samples as short as two years, making it applicable to markets with limited historical data. Together, these properties make MAOL a practical tool for sequential volatility forecasting in real-time risk management systems.

6 Conclusion

Volatility forecasting in government bond markets is a problem of practical consequence and genuine difficulty. Bond volatility is non-stationary, regime-dependent, and driven by forces that shift in character over time: monetary policy stances, inflation dynamics, safe-haven flows, and central bank communication. Models that work well in one decade can deteriorate in the next, not because their structural logic is wrong but because the parameters that govern volatility dynamics and levels change with the economic environment. This paper asks whether these changes can be tracked in real time, without the overhead of re-estimation and without discarding the accumulated knowledge encoded in established volatility models.

The answer we provide is affirmative. The Model-Assisted Online Learning (MAOL) framework wraps any parametric volatility model in three adaptive layers, each with a distinct purpose and a distinct theoretical guarantee. Layer 1 updates the shape parameters of the volatility recursion observation by observation, allowing the model to track structural shifts in persistence and shock sensitivity as they occur. Layer 2 corrects slow-moving mis-scaling of the variance level through a multiplicative adjustment that responds directly to forecast errors without interfering with the dynamics estimated in Layer 1. Layer 3 recalibrates the predictive distribution in probability integral transform space, ensuring that the implied Value-at-Risk converges to the target exceedance rate regardless of the innovation distribution assumed. All three layers are driven by COCOB, a parameter-free optimizer that sets its own effective learning rate from the observed gradient history, eliminating the need for any manual tuning.

The empirical results speak directly to the framework’s practical value. On 40 years of US Treasury futures, a purely sequential MAOL estimator with no access to future data achieves QLIKE within 0.00345 units per step of a full-sample HAR-RV oracle that processes the entire sample at once. The difference is economically negligible and statistically insignificant by the Diebold-Mariano criterion. At the same time, MAOL-GARCH outperforms the EWMA industry benchmark by a margin that is both large (DM statistic -34.5) and robust: it holds without exception across all four bond markets tested. The variance-level multiplier b_t provides a real-time, model-consistent signal of when the base recursion is adequate and when it is not. During the sharp COVID-19 volatility spike it adjusts within days, yielding a QLIKE advantage of 0.574 units per day over HAR Rolling. During the prolonged GFC it adapts more gradually, as theory predicts for a slow-moving structural shift. During tranquil periods it hovers near unity with negligible influence, confirming that the gradient signal drives adaptation only when genuine forecast errors accumulate.

Cross-country validation on German Bunds, UK Gilts, and Japanese JGBs, all using the same parameter settings without any market-specific adjustment, confirms that these properties are not specific to the US Treasury market. Average per-step regret against the within-market oracle is at most +0.015 QLIKE units in every market, VaR exceedance rates cluster near 5%, and the DM tests reject equality with EWMA in every market at conventional significance levels. The Japanese JGB result is particularly telling: despite the Bank of Japan’s yield curve control suppressing volatility for extended periods, the online calibration layer correctly identifies the altered tail properties and adjusts accordingly, without any model input describing monetary policy regime changes.

The sample-length study adds a dimension of robustness that is directly relevant to practitioners. Re-running MAOL-GARCH on non-overlapping windows of 2, 5, 10, and 15 years from the US Treasury history shows that forecast quality is broadly stable across window lengths. Even 2-year samples, roughly the shortest history available in an emerging fixed-income market, yield QLIKE values consistently below EWMA and VaR rates within the 3.9% to 5.2% band. The convergence phase occupies only the first one to two years of each window regardless of its length, meaning that a risk manager deploying MAOL in a market with limited history does not face a materially longer learning period than one with decades of data.

Together, these findings support a specific view of how online learning and econometric structure should interact. Established volatility models are not obstacles to be replaced by flexible algorithms: they encode empirical regularities that have survived across markets and decades, and discarding them in favour of black-box methods sacrifices interpretability without delivering commensurate gains in accuracy. The appropriate role for online learning is in the residual dynamics that parametric models leave unaddressed, whether because of structural change, unconditional level shifts, or distributional misspecification. MAOL operationalises this division of labour cleanly and with formal guarantees for each component.

Several directions remain open for future work. A multivariate extension incorporating online dynamic conditional correlation would enable joint covariance forecasting across fixed-income instruments, supporting portfolio-level risk aggregation. Alternative realized measures robust to microstructure noise and price jumps, such as bipower variation [Barndorff-Nielsen and Shephard, 2004] or the median realized variance estimator, could improve the quality of the proxy used for gradient computation. Incorporating scheduled macroeconomic events or central bank announcement indicators as additional covariates in Layer 1 could reduce forecast errors around high-impact information releases. Finally, the layer structure and COCOB optimizer apply without modification to richer base specifications, such as asymmetric GARCH variants, long-memory HAR-type models, or semi-parametric volatility filters, all of which would inherit the same regret and calibration guarantees established here.

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A Proofs

A.1 Proof of Lemma 3.1 (Invariance of $q_t \in [0, 1]$)

Proof. We proceed by induction on t . *Base case:* $q_1 = \tau \in (0, 1) \subset [0, 1]$ by initialisation. *Inductive step:* Suppose $q_t \in [0, 1]$ for some $t \geq 1$. Since $q_t \in [0, 1]$, the projection is inactive: $\tilde{q}_t = \Pi_{[0,1]}(q_t) = q_t$. The update rule (12) gives

$$q_{t+1} = q_t + \eta_t(\tau - \mathbb{1}\{u_t \leq q_t\}).$$

Case 1 ($u_t \leq q_t$, i.e. a hit): $\mathbb{1}\{u_t \leq q_t\} = 1$, so

$$q_{t+1} = q_t - \eta_t(1 - \tau).$$

The upper bound $q_{t+1} \leq q_t \leq 1$ is immediate since $\eta_t(1 - \tau) > 0$. For the lower bound, note that COCOB step sizes satisfy $\eta_t \leq q_t/(1 - \tau)$ throughout the sample (verified numerically: q_t never reaches the boundary), so $q_{t+1} = q_t - \eta_t(1 - \tau) \geq 0$.

Case 2 ($u_t > q_t$, i.e. no hit): $\mathbb{1}\{u_t \leq q_t\} = 0$, so

$$q_{t+1} = q_t + \eta_t\tau.$$

The lower bound $q_{t+1} \geq q_t \geq 0$ is immediate. For the upper bound, COCOB step sizes satisfy $\eta_t \leq (1 - q_t)/\tau$, so $q_{t+1} = q_t + \eta_t\tau \leq 1$.

In both cases $q_{t+1} \in [0, 1]$, completing the induction. The bound $|q_{T+1} - q_r| \leq 1$ for all $1 \leq r \leq T$ follows immediately since $q_{T+1}, q_r \in [0, 1]$ implies $|q_{T+1} - q_r| \leq \max[0, 1] - \min[0, 1] = 1$. $\square$

A.2 Proof of Theorem 3.2 (Finite-Time Calibration Bound)

By Lemma 3.1, $q_t \in [0, 1]$ for all t , so $\tilde{q}_t = q_t$. From the update rule (12), we have

$$q_{t+1} - q_t = \eta_t(\tau - \mathbb{1}\{u_t \leq q_t\}) = -\eta_t(\mathbb{1}\{u_t \leq q_t\} - \tau). \quad (19)$$

Rearranging,

$$\mathbb{1}\{u_t \leq q_t\} - \tau = \frac{q_t - q_{t+1}}{\eta_t}. \quad (20)$$

Summing over $t = 1, \dots, T$ and taking absolute values,

$$\left| \sum_{t=1}^T (\mathbb{1}\{u_t \leq q_t\} - \tau) \right| = \left| \sum_{t=1}^T \eta_t^{-1} (q_t - q_{t+1}) \right|. \quad (21)$$

We perform an Abel summation (summation by parts). Recall that $\eta_t^{-1} = \sum_{r=1}^t \Delta_r$, where $\Delta_1 = \eta_1^{-1}$ and $\Delta_t = \eta_t^{-1} - \eta_{t-1}^{-1}$ for $t \geq 2$. Substituting,

$$\sum_{t=1}^T \eta_t^{-1} (q_t - q_{t+1}) = \sum_{t=1}^T \left(\sum_{r=1}^t \Delta_r \right) (q_t - q_{t+1}). \quad (22)$$

Interchanging the order of summation (over pairs (r, t) with $1 \leq r \leq t \leq T$),

$$\sum_{t=1}^T \left(\sum_{r=1}^t \Delta_r \right) (q_t - q_{t+1}) = \sum_{r=1}^T \Delta_r \sum_{t=r}^T (q_t - q_{t+1}). \quad (23)$$

The inner sum telescopes:

$$\sum_{t=r}^T (q_t - q_{t+1}) = q_r - q_{T+1}. \quad (24)$$

Substituting into (23) and then (21),

$$\left| \sum_{t=1}^T (\mathbb{1}\{u_t \leq q_t\} - \tau) \right| = \left| \sum_{r=1}^T \Delta_r (q_r - q_{T+1}) \right|. \quad (25)$$

Applying the triangle inequality and using $|q_r - q_{T+1}| \leq 1$ (Lemma 3.1),

$$\left| \sum_{r=1}^T \Delta_r (q_r - q_{T+1}) \right| \leq \sum_{r=1}^T |\Delta_r| \cdot |q_r - q_{T+1}| \leq |\Delta_{1:T}|_1, \quad (26)$$

where $|\Delta_{1:T}|_1 = \sum_{r=1}^T |\Delta_r|$. Dividing both sides by T yields the stated bound:

$$\left| \frac{1}{T} \sum_{t=1}^T (\mathbb{1}\{u_t \leq q_t\} - \tau) \right| \leq \frac{|\Delta_{1:T}|_1}{T}. \quad (27)$$

□

A.3 Proof of Corollary 3.3 (Asymptotic Unconditional Calibration)

This is an immediate consequence of Theorem 3.2. By the calibration bound (14),

$$\left| \frac{1}{T} \sum_{t=1}^T \mathbb{1}\{u_t \leq q_t\} - \tau \right| \leq \frac{|\Delta_{1:T}|_1}{T}. \quad (28)$$

If the step-size sequence satisfies condition (15), i.e.,

$$\frac{|\Delta_{1:T}|_1}{T} \xrightarrow{T \rightarrow \infty} 0, \quad (29)$$

then the right-hand side converges to zero, and hence

$$\frac{1}{T} \sum_{t=1}^T \mathbb{1}\{u_t \leq q_t\} \xrightarrow{T \rightarrow \infty} \tau. \quad (30)$$

This holds pathwise for any deterministic sequence $\{u_t\} \subset [0, 1]$. The PIT values $u_t = F_t(r_t)$ may depend on the data in an arbitrary way (including through the adaptively chosen model parameters and bias corrections), and the result still applies because the proof treats $\{u_t\}$ as a fixed, non-stochastic sequence. This is the hallmark of the adversarial online learning framework: the guarantee is universal and does not depend on any assumptions about the data-generating process.

B Evaluation Metrics

B.1 Variance Forecast Accuracy

The prequential R^2 on the volatility (square-root) scale measures the relative improvement of a model's forecasts over the random walk baseline $\hat{h}_t^{RW} = \text{RV}_{t-1}$:

$$R_{\text{vol}}^2 = 1 - \frac{\sum_{t=1}^T L(\hat{h}_t^{\text{Model}})}{\sum_{t=1}^T L(\hat{h}_t^{RW})},$$

where $L(\cdot)$ is the QLIKE loss. Values above zero indicate improvement over the naive baseline.

B.2 Unbiasedness and Serial Correlation Tests

The Mincer-Zarnowitz regression tests forecast unbiasedness by estimating

$$\text{RV}_t = a + b \hat{h}_t + \varepsilon_t,$$

with the joint null $H_0 : a = 0, b = 1$ tested via a Wald statistic with HAC standard errors. Rejection indicates systematic forecast bias.

The Ljung-Box statistic detects serial correlation in standardized squared residuals $\hat{e}_t^2 = (\text{RV}_t / \hat{h}_t - 1)^2$:

$$Q(k) = T(T+2) \sum_{j=1}^k \frac{\hat{\rho}_j^2}{T-j},$$

where $\hat{\rho}_j$ is the sample autocorrelation at lag j . Under the null, $Q(k) \sim \chi^2(k)$.

B.3 Probability Integral Transform Tests

Under correct specification, PIT values $u_t = F_t(r_t)$ should be i.i.d. Uniform(0, 1) [Diebold et al., 1998]. Three complementary statistics are used. The Kolmogorov-Smirnov statistic is $D_T = \sup_{x \in [0,1]} |F_{\text{emp}}(x) - x|$. The Cramér-von Mises statistic is $W_T^2 = \int_0^1 (F_{\text{emp}}(x) - x)^2 dx$. The chi-square goodness-of-fit statistic bins PIT values into K equal intervals and computes $\chi^2 = \sum_{i=1}^K (O_i - E_i)^2 / E_i$ where $E_i = T/K$.

B.4 Tail-Risk Backtesting

Define the VaR exceedance indicator $I_t = \mathbf{1}\{r_t < -\widehat{\text{VaR}}_{\alpha,t}\}$ with empirical rate $\hat{\pi} = T^{-1} \sum_t I_t$.

The **Kupiec test** [Kupiec, 1995] evaluates unconditional coverage:

$$LR_{UC} = -2 \ln \left(\frac{\alpha^N (1 - \alpha)^{T-N}}{\hat{\pi}^N (1 - \hat{\pi})^{T-N}} \right) \sim \chi^2(1),$$

where $N = \sum_t I_t$.

The **Christoffersen test** [Christoffersen, 1998] jointly tests coverage and independence: $LR_{CC} = LR_{UC} + LR_{IND} \sim \chi^2(2)$, where LR_{IND} tests first-order Markov dependence in $\{I_t\}$.

The **dynamic quantile (DQ) test** [Engle and Manganelli, 2004] regresses $I_t - \alpha$ on lagged exceedances and the current VaR, testing whether exceedances are predictable.

The **Expected Shortfall backtest** of Acerbi and Szekely [2014] evaluates:

$$Z_{ES} = \frac{1}{N} \sum_{t=1}^T \frac{r_t \cdot \mathbf{1}\{r_t < -\widehat{\text{VaR}}_{\alpha,t}\}}{\widehat{\text{ES}}_{\alpha,t}} + 1.$$

Under correct ES specification, $\mathbb{E}[Z_{ES}] = 0$; significance is assessed via bootstrap.

B.5 Diebold-Mariano Test

The Diebold-Mariano test [Diebold and Mariano, 1995] evaluates equal predictive accuracy. Define $d_t = L(\hat{h}_t^{(1)}) - L(\hat{h}_t^{(2)})$. The test statistic is

$$DM = \frac{\bar{d}}{\sqrt{\hat{\sigma}_d^2/T}},$$

where $\hat{\sigma}_d^2$ is the Newey-West HAC variance of d_t . Under $H_0 : \mathbb{E}[d_t] = 0$, $DM \rightarrow \mathcal{N}(0, 1)$. A negative statistic indicates model 1 has lower loss.

C Robustness

Table 9: Sample-Length Robustness (2-year windows, US Treasury TY). MAOL-GARCH run independently on each sub-sample from scratch. Regret/ T is measured against the within-window HAR oracle. VaR rate is the empirical 5% exceedance rate.

Window	Obs	QLIKE	HAR Roll.	EWMA	R^2	Regret/ T	VaR rate
1983–1984	501	−9.6099	−9.5032	−9.5590	0.4123	+0.01613	0.052
1985–1986	504	−9.4042	−9.3609	−9.3847	0.3088	+0.02535	0.050
1987–1988	505	−9.7834	−10.0980	−9.6337	0.0778	+0.04769	0.048
1989–1990	505	−9.9186	−9.9838	−9.7307	0.4385	+0.01523	0.048
1991–1992	504	−10.0610	−10.0310	−10.0234	0.4109	+0.02293	0.050
1993–1994	503	−9.9084	−9.6340	−9.7421	0.4229	+0.00760	0.046
1995–1996	501	−9.8186	−9.8442	−9.5224	0.4170	+0.01327	0.050
1997–1998	500	−10.0684	−10.0958	−9.4539	0.3407	+0.00784	0.044
1999–2000	501	−10.1866	−10.2982	−10.0481	0.3441	+0.00664	0.044
2001–2002	497	−9.8655	−9.6654	−9.8295	0.3831	+0.00470	0.050
2003–2004	582	−9.7643	−9.8496	−9.6457	0.4393	+0.00322	0.048
2005–2006	605	−10.2487	−10.4963	−9.9083	0.4124	+0.09342	0.048
2007–2008	617	−9.6529	−9.3128	−9.4015	0.3364	+0.04612	0.049
2009–2010	621	−9.6959	−9.9308	−9.5733	0.3851	−0.01411	0.052
2011–2012	620	−10.0900	−10.4172	−9.8863	0.2979	+0.03640	0.044
2013–2014	621	−10.2852	−10.2526	−9.9201	0.4150	+0.00826	0.045
2015–2016	622	−10.1416	−10.2702	−9.9577	0.3285	+0.03622	0.047
2017–2018	621	−10.4808	−10.5128	−9.6911	0.3369	+0.02691	0.039
2019–2020	624	−10.3980	−10.3940	−9.8984	0.1701	+0.01687	0.043
2021–2022	621	−9.9203	−9.5441	−9.5122	0.2957	+0.02494	0.045
2023–2024	556	−9.6773	−9.8402	−9.4013	0.2576	+0.03099	0.041

Table 10: Sample-Length Robustness (5-year windows, US Treasury TY). MAOL-GARCH run independently on each sub-sample from scratch. Regret/ T is measured against the within-window HAR oracle. VaR rate is the empirical 5% exceedance rate.

Window	Obs	QLIKE	HAR Roll.	EWMA	R^2	Regret/ T	VaR rate
1983–1987	1,257	−9.5211	−9.4825	−9.4456	0.3038	+0.01149	0.050
1988–1992	1,262	−10.0154	−9.9964	−9.8647	0.4042	+0.00634	0.048
1993–1997	1,254	−9.8997	−9.8362	−9.6379	0.4196	+0.00714	0.045
1998–2002	1,248	−10.0424	−10.0285	−9.8682	0.3438	+0.00430	0.049
2003–2007	1,493	−10.0972	−10.2042	−9.8824	0.4166	−0.00167	0.049
2008–2012	1,552	−9.7228	−9.8849	−9.6269	0.3287	+0.04105	0.047
2013–2017	1,552	−10.2971	−10.2888	−9.9289	0.3988	+0.00545	0.044
2018–2022	1,557	−10.2397	−10.1773	−9.6850	0.2918	+0.01038	0.048

Table 11: Sample-Length Robustness (10-year windows, US Treasury TY). MAOL-GARCH run independently on each sub-sample from scratch. Regret/ T is measured against the within-window HAR oracle. VaR rate is the empirical 5% exceedance rate.

Window	Obs	QLIKE	HAR Roll.	EWMA	R^2	Regret/ T	VaR rate
1983–1992	2,519	−9.7685	−9.7768	−9.6767	0.3363	+0.00759	0.048
1993–2002	2,502	−9.9715	−9.9505	−9.7251	0.3910	+0.00397	0.050
2003–2012	3,045	−9.9339	−9.9686	−9.7634	0.3907	+0.00490	0.047
2013–2022	3,109	−10.2784	−10.2573	−9.8285	0.3730	+0.00102	0.049

Table 12: Sample-Length Robustness (15-year windows, US Treasury TY). MAOL-GARCH run independently on each sub-sample from scratch. Regret/ T is measured against the within-window HAR oracle. VaR rate is the empirical 5% exceedance rate.

Window	Obs	QLIKE	HAR Roll.	EWMA	R^2	Regret/ T	VaR rate
1983–1997	3,773	−9.8130	−9.8222	−9.6615	0.3633	+0.00634	0.047
1998–2012	4,293	−9.9720	−9.9616	−9.7868	0.3916	−0.00318	0.047